



Research review paper

Plant synthetic biology for producing potent phyto-antimicrobials to combat antimicrobial resistance

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ABSTRACT

Inappropriate and injudicious use of antimicrobial drugs in human health, hygiene, agriculture, animal husbandry and food industries has contributed significantly to rapid emergence and persistence of antimicrobial resistance (AMR), one of the serious global public health threats. The crisis of AMR *versus* slower discovery of newer antibiotics put forth a daunting task to control these drug-resistant *superbugs*. Several phyto-antimicrobials have been identified in recent years with direct-killing (bactericidal) and/or drug-resistance reversal (re-sensitization of AMR phenotypes) potencies. Phyto-antimicrobials may hold the key in combating AMR owing to their abilities to target major microbial drug-resistance determinants including cell membrane, drug-efflux pumps, cell communication and biofilms. However, limited distribution, low intracellular concentrations, eco-geographical variations, beside other considerations like dynamic environments, climate change and over-exploitation of plant-resources are major blockades in full potential exploration phyto-antimicrobials. Synthetic biology (SynBio) strategies integrating metabolic engineering, RNA-interference, genome editing/engineering and/or systems biology approaches using plant chassis (as *engineerable* platforms) offer prospective tools for production of phyto-antimicrobials. With expanding SynBio toolkit, successful attempts towards introduction of entire gene cluster, reconstituting the metabolic pathway or transferring an entire metabolic (or synthetic) pathway into heterologous plant systems highlight the potential of this field. Through this perspective review, we are presenting herein the current situation and options for addressing AMR, emphasizing on the significance of phyto-antimicrobials in this apparently *post-antibiotic* era, and effective use of plant chassis for phyto-antimicrobial production at industrial scales along with major SynBio tools and useful databases. Current knowledge, recent success stories, associated challenges and prospects of translational success are also discussed.

1. Introduction

The global rise of antimicrobial resistance (AMR) poses an emerging threat to human health and environment, attributed to the emergence and persistence of drug-resistant bacterial phenotypes, often referred to as the “superbugs”. The extent of AMR has been reported at three rising levels- the multidrug resistance (MDR), extensive drug resistance (XDR) and the pan-drug resistance (PDR) (Shriram et al., 2018). The MDR can be defined as the acquisition of non-susceptibility to at least one agent in three or more antimicrobial classes, XDR means non-susceptibility to at

least one agent in all, except two or fewer antimicrobial categories, whereas non-susceptibility to all antimicrobial agents from all available classes is known as PDR (Exner et al., 2017; Spengler et al., 2017). Owing to its threatening nature, AMR is rightly declared as a *global risk* by the World Economic Forum (WEF, 2013). Since the initial reports of drug resistance in *Enterobacteria* in the 1950s, large number of strains with AMR phenotypes are being reported and unfortunately their number as well as the resistance level is on rise (Levy, 2001). The growing concern of AMR in microbes put forth a key challenge, surpassing the availability of newer drugs and indiscriminate use of the

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existing ones.

A major threat is being posed by 'ESKAPE' pathogens, a group of threatening drug-resistant pathogens including *Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa* and *Enterobacter* species (Yu et al., 2020). The members of this group have been reported as MDR, XDR and PDR phenotypes (Mapara et al., 2015; Thomas et al., 2019), and are designated as World Health Organization (WHO) Critical Priority I and II pathogens (WHO, 2017), rendering even the most potent drugs ineffective. The ESKAPE pathogens are common causative agents for the nosocomial infections amid the patient with critical illness or with the compromised immunity. This group symbolizes the new paradigms in bacterial pathogenesis, transmission as well as the AMR.

The emergence of new resistance mechanisms in microbes contribute to inefficacy of the existing drugs, prolonged illness and escalating expenditures (WHO, 2018). This condition worsens in immunocompromised patients (as in diabetes, organ-transplantations and HIV-infection) and the subsequent susceptibility of a patient to infections leads to further AMR incidences (Tanwar et al., 2014). The global rise in the frequency, magnitude, spread and persistence of drug resistance phenotypes in individual microbes has significantly contributed to the challenge. An estimate suggested that 90% strains of *Staphylococcus aureus* were penicillin resistant globally and approximately 70% showed resistance to methicillin (Hemaiswarya et al., 2008). The rising statistics of AMR infections is estimated to result in 10 million deaths per year by 2050, with Asia and Africa accounting for 4.7 million and 4.1 million deaths, respectively (O'Neill, 2014). Though AMR occurs naturally in microbes, the injudicious use of commonly available antibiotics in human health, hygiene, agriculture, animal husbandry and food industries speed-ups the emergence of drug-resistant strains (WHO, 2014; Yu et al., 2020). The AMR once gained by a bacterial strain may successively be transmitted to the next progeny via vertical gene transfer or to other bacteria through horizontal gene transfer, making their treatment difficult (B. Li et al., 2019; Li et al., 2020; Wang et al., 2021).

To address the challenges, discovery of new antimicrobials defines a global research priority; however, only few new classes of antibiotics, like the cyclic lipopeptides and oxazolidinones, have been marketed for treating Gram-positive bacterial infections (WHO, 2011). The problem of AMR is more prevalent and severe in Gram-negative bacteria, owing mainly to their outer membranes (Yu et al., 2020). The outer membrane (cell-wall) in Gram-negative bacteria serves as a permeability barrier and exclude drugs from penetrating the cell and thus requiring high antibiotic doses with possible toxicity issues (Exner et al., 2017; Masi et al., 2017; Theuretzbacher et al., 2020). According to a recent report of WHO, there is a worrying gap between highly drug-resistant Gram-negative pathogens and the available options for treating them (WHO, 2020). In the present era, decline in innovative approaches in antibiotic development and decreasing investments threaten the global initiative to counter drug-resistant infections. The Global Antibiotic Research and Development Partnership (GARDP), a non-profit organization of WHO and the Drugs for Neglected Diseases Initiative (DNDi) were established to promote the development of new antibiotics (with improved efficacy) to combat rising microbial drug-resistance.

The growing need to discover new antimicrobials have encouraged the exploration of more and more medicinal plant resources for targeting and restricting bacterial drug-resistance. Plant natural products (PNPs) have shown their abilities in effectively targeting the major drug-resistance mechanisms reported in bacteria including membranes (Côté et al., 2019) and membrane-proteins (Suarez et al., 2005), biofilms (Abraham et al., 2011), drug efflux pumps (Aghayan et al., 2017) and bacterial cell communication (quorum sensing) (Jakobsen et al., 2012; Borges et al., 2014). Further, a promising approach accounts for the improvement of clinical efficacy of existing antibiotics by rejuvenating them with the help of bioactive molecules (as adjuvants), an approach known as re-sensitization or re-potentialization of existing but non-effective antibiotics (Borges et al., 2016; Shriram et al., 2018; Yu et al., 2020).

In this context, SynBio is seen to revolutionize and redirect the discovery of "botanical therapeutics", a prospective alternative source to tackle rising AMR through allowing large-scale production and sustainable supply of effective phyto-antimicrobials. SynBio presents an approach that involves redesigning organisms for useful purposes by engineering them to have new abilities. Recent advances in SynBio focus on addressing the challenges associated with phytochemicals and their respective metabolic pathways, aiding in synthetic pathway reconstitution and increased production in heterologous systems. Exploration of biosynthetic potential of biological genomes for production of PNPs may reinvigorate the drying antibiotic pipelines and opening new avenues to novel therapeutic entities. The emerging trends in SynBio and production of natural products showcase focus on creation of genetic parts; for example, promoters for precise gene expression and activity prediction through construction of *in silico* models, high-throughput DNA construction and assembly approaches, computer-aided design tools to improve prediction of future experiments and designing biological genomes via genome editing (Smanski et al., 2016). On the other hand, metabolic engineering approaches seek for the optimization of cellular processes through engineering or manipulations of metabolic pathway (s), endemic to a specific organism, to produce specific metabolites (García-Granados et al., 2019). Metabolic engineering and SynBio, both fields often depend on each other rather than working separately to achieve their goals (García-Granados et al., 2019).

Synthetic manipulations of microbial hosts have made good strides as sound platforms for production of specialized metabolites and inclusion of plant specialized pathways in microbial systems (Galanie et al., 2015). However, there are challenges associated with the use of microbial hosts for PNP production including the availability of precursors for the biosynthesis of PNPs, inefficient activities of some plant enzymes in microbes, and low output of metabolic fluxes in microbes (Li et al., 2018a, 2018b), which necessitates exploring alternative production platforms for production of PNPs. Plant chassis as platforms for production of PNPs offer advantages over their microbial counterparts, as discussed in Sections 5.1 and 5.2.

Additionally, the use of algal systems for production of high-value metabolites implementing SynBio strategies is gaining momentum (Moses et al., 2017; Lauersen, 2019). The unicellular organisms like microalgae and cyanobacteria are emerging chassis/platforms for production of high-value compounds, attributed to their rapid multiplication rate, light-driven growth and unicellular physiology. For instance, algal engineering has been attempted for the isoprenoid production (Lauersen, 2019). The attempts have been made towards algal engineering focused on tailoring metabolism and metabolic pathway engineering for production of high-value metabolites, including 2,3-Butanediol in *Synechococcus* sp. PCC 7942 (Oliver et al., 2013), Limonene in *Synechococcus* sp. PCC 7002 (Davies et al., 2014), Carotenoids in *Chlamydomonas reinhardtii* (Couso et al., 2011), Isoprene in *Synechococcus* sp. PCC 7942 (Gao et al., 2016) among others. However, despite the development of several transformation tools and approaches, the low efficiency of foreign gene expression in algal systems is still a challenge (Doron et al., 2016). Besides, chloroplast transformation toolkits have been established for several microalgae, but similar tools for nuclear-based protein expression remain under-developed, as for land plants (Doron et al., 2016). One more bottleneck in the full-potential exploration of algal systems is the lack of searchable algal phenomics databases (Fabris et al., 2020).

Though, a long way to go, plant chassis as scalable platforms for production of high-value PNPs are emerging as self-sufficient systems deriving carbon supply and energy from photosynthesis beside other advantages. As engineerable platforms, plant systems have shown significant success in production of antimicrobial peptides (AMPs) (Chahardoli et al., 2018). Therefore, SynBio strategies integrating metabolic engineering, RNA interference (RNAi), genome engineering and systems biology approaches using plant chassis for production of phyto-antimicrobials production and its optimization defines the prospective

tools to solve growing concern of AMR. Through this perspective review, we are presenting herein the current scenario of threatening AMR, possible options with us for combating it highlighting the significance of natural products in the modern “post-antibiotic” era, and critically assessed if and how plant cell factories and their chassis can be sustainably used for phyto-antimicrobial production at industrial scales. Current knowledge and recent success stories, associated challenges and prospects of translational success are also discussed.

2. Emerging resistance against conventional antimicrobial agents

Persistent and injudicious application of antibiotics has led to the

unfortunate development of severe AMR in increasing number of bacterial strains, leaving the most potent drugs ineffective, not only in nosocomial but in community environments as well (Mulani et al., 2019). Fundamental mechanisms (replication, transcription, translation) in bacteria have historically been major targets for most of the antibiotics in use. However, in a quest for survival and their supremacy over these antibiotics, bacteria have developed several mechanisms such as changes in the membrane permeability, enzymatic inactivation of antibiotics, and modifications of these antibiotic targets.

2.1. Mechanistic insights into the action of antimicrobial agents

Antimicrobial drugs (or antibiotics), both synthetic and of natural

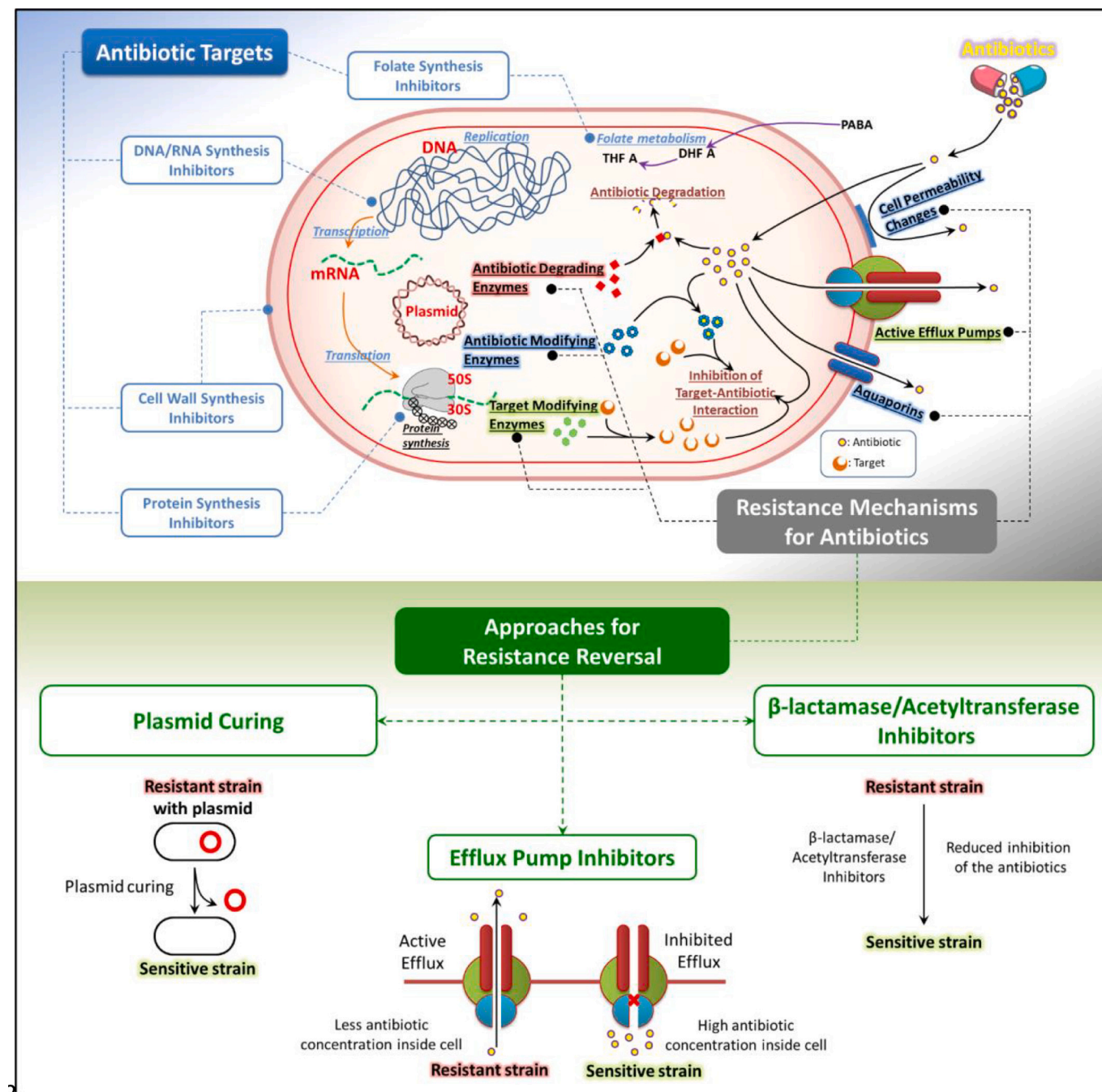


Fig. 1. Targets of different antibiotics (blue), mechanisms that lead to the development of antibiotic resistance in bacteria (grey) and approaches to reverse antibiotic resistance (green). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

origin, target key microbial characters and core biological mechanisms/processes, refer to Fig. 1. Bacterial cell wall with polymer peptidoglycans as its basic constituents is a vital polysaccharide assembly surrounding bacterial cell and keeps the cytoplasmic membrane protected and intact (Cho et al., 2014a). Targeting the biogenesis of the cell wall is one of the most prominent strategies for developing effective drugs/antibiotics. Antibiotics inhibit the functioning of penicillin-binding proteins or PBPs (trans-peptidation: crosslinking; trans-glycosylation: polymerization of the glycan strands). The β -lactam antibiotics target cell wall biogenesis via inducing drug-mediated imbalance between the peptidoglycan synthases and peptidoglycan hydrolases activities (Cho et al., 2014a). Apart from the β -lactams, some non-lactam antibiotics such as glycopeptides also target the cell wall biosynthesis as their mode of action. Glycopeptides bind with C-terminal D-Ala-D-Ala residues of the lipid attached peptidoglycan precursors via five H-bonds, shielding the D-Ala-D-Ala site from the transpeptidation (Sarkar and Haldar, 2019). Colistin binds with lipopolysaccharides and the phospholipids in the outer bacterial cell membrane and displaces the divalent cations (Ca^{2+} and Mg^{2+}) which results in cell-membrane rupture (Biswas et al., 2012).

Other targets include bacterial ribonucleoprotein and the translational machinery (Arenz and Wilson, 2016). There are three major drug targets on bacterial ribosomes, first one being the A-site on 30S subunit, where drug hinders the codon recognition and translocation process. Second target is peptidyl transferase centre on 50S, where interaction of this center with antibiotics inhibits the formation of peptide bonds, and the third is peptide exit tunnel on 50S ribonucleoprotein subunit, where the antibiotics block the growth of amino acid chain (McCoy et al., 2011). Antibiotics that inhibit the protein synthesis include aminoglycosides, tetracyclines, macrolides, streptogramins, oxazolidinones, chloramphenicol and lincosamides (Bhattacharjee, 2016). Beside these translation-associated targets, some antibiotics target enzymes involved in the DNA replication, for example quinolones and their derivative fluoroquinolones hamper the activities of bacterial topoisomerases (DNA gyrase and topoisomerase IV) (Redgrave et al., 2014) which are vital for DNA replication, transcription, recombination and repair processes. Hindrance in the activities of these enzymes usually leads to the arrest of DNA replication fork progression and subsequent cell death (Hernández et al., 2011). On the similar line, nitroimidazole antibiotics such as metronidazole are known to induce DNA damage and cell death (Ang et al., 2017).

Apart from these drugs targeting the core bacterial processes, some classes of antibiotics target other essential pathways such as folate synthesis (Sheldon, 2005). Antibiotics from the class sulfonamides or dihydrofolate reductase inhibitors like trimethoprim, inhibits the enzymes dihydropteroate synthase and dihydrofolate reductase, respectively, the key enzymes of folic acid synthetic pathway (Kapoor et al., 2017). There are some new antibiotics going through various phases of clinical trials with promising results (Hutchings et al., 2019; Butler and Paterson, 2020). For instance, Sulopenem, a member from the class penem, is under phase III clinical trials, which targets PBPs. The clinical trials are focused on the urinary tract infections targeting the resistant Enterobacteriaceae (Dunne et al., 2020). Similarly, phase III clinical trials of anti-MRSA (methicillin resistant *Staphylococcus aureus*) agent Levonadifloxacin (Fluoroquinolone, targeting DNA gyrase and topoisomerase) have also been completed (Bhatia et al., 2020). However, these efforts are threatened by the rapid rise of AMR against every clinically applied antibiotic, necessitating novel strategies to resolve this threatening public health concern (Ragheb et al., 2019; Yu et al., 2020).

2.2. Mechanisms of antimicrobial resistance and spread

Bacteria exposed to antimicrobials tend to develop intrinsic AMR via inherent structural or functional characteristics such as absence of susceptible antibiotic target or difference in the outer membrane composition. However, AMR is also acquired through several other

mechanisms as given in Fig. 1, such as (1) minimizing the intracellular antibiotic concentrations via poor penetration or active efflux, (2) modifying the antibiotic targets via mutations or post-translational modifications, and (3) inactivating the antibiotics via modification/hydrolysis (reviewed by Blair et al., 2015). In the first mechanism, the target accession to the antibiotics is prevented via reducing the cell permeability and thus reducing the intracellular antibiotic concentration. This is accomplished by down-regulation of the porin proteins, which act as the diffusion channels for entry of hydrophilic antibiotics or by substituting the porins with more selective channels (Tamber and Hancock, 2003). On the other hand, active efflux of the antibiotics via action of bacterial efflux pumps is another way to reduce the intracellular antibiotic concentrations (Fig. 1). The efflux pumps have a remarkable capability to eliminate toxic substances out of the bacterial cell and are vital for maintenance of the pathogenicity. Hence increased expression of the efflux pump corresponds to an increased level of AMR (Shriram et al., 2018). Based on their energy source and specificities, bacterial efflux pumps are classified into (1) ATP-binding cassette (ABC) multidrug transporters, and (2) proton motive force (PMF) mediated secondary transporters (Putman et al., 2000). The PMF family is further classified into four sub-classes, namely; resistance-modulation-cell division (RND), multidrug and toxic compound extrusion (MATE), the major facilitator superfamily (MFS), and small-MDR (SMR) family (Sun et al., 2014). Efflux pumps are a vital mode in gaining AMR due to their capacity to exclude most of the unwanted substances (Pidcock, 2014; Venter et al., 2015).

Modification of the drug targets is another strategy acquired by bacteria for developing AMR characters (Fig. 1), where structural changes in the targets prevent the optimal antibiotic-target interaction, however at the same time enable the targets to carry out their normal function. This can be achieved via single point mutations in the genes encoding the antibiotic targets, which results in the lessened affinity for an antibiotic. Similarly, chemical modification of targets mediated by the addition of chemical groups (for instance, the methylation of ribosomal sites acting as antibiotic targets) also results in avoidance of drug-binding without compromising the normal target activity, refer to Fig. 1 (Fritsche et al., 2008; Shen et al., 2013). Besides, bacterial pathogens can directly aim for the destruction or inactivation of antibiotics with the help of enzymes like carbapenemases, β -lactamases and extended-spectrum β -lactamases (ESBLs) (French et al., 2017). Several enzymes from these classes have been identified that inactivate the antibiotics through hydrolysis (Munita and Arias, 2016). On the other hand, bacterial enzymes can also modify the antibiotics via addition of chemical groups (such as acyl, nucleotidyl, phosphate, and ribityl groups) to the vulnerable sites on antibiotic molecule creating a steric hindrance and prevention of antibiotic-target interaction (Wright, 2005).

The vital cellular functions are prominent drug-targets, hence induction of the stress responses in bacterial cells to combat the impacts of antibiotics or to acquire resistance against them are predictable. Such stress interactions include SOS responses and rpoS regulons (Baharoglu and Mazel, 2011; Thi et al., 2011; Gutierrez et al., 2013). The presence of single-stranded DNA produced because of the DNA damage or DNA replication arrest persuades the SOS response which is facilitated by the auto-cleavage of the LexA repressor. The LexA cleavage stimulates the expression of several genes associated with DNA damage (Courcelle et al., 2001; Friedberg et al., 2006). The bacterial SOS response can be correlated with either action of antibiotics (quinolones) targeting enzymes interacting with DNA (Drlica and Zhao, 1997), two-component systems like DpiAB (β -lactams) (Miller et al., 2004), or stimulated production of reactive oxygen species (aminoglycosides) (Kohanski et al., 2007, 2008). The bactericidal action of the antibiotics also acts as a stimulus for the general stress response, which is regulated by RpoS sigma factor (Gutierrez et al., 2013). This is a stationary phase sigma factor which is induced in exponential phase against stress (Merrikh et al., 2009; Battesti et al., 2012). Following the induction of this regulon, the genes coding for the defense enzymes like catalases or ion-

chelators get induced to protect the cell from oxidative damage (Barth et al., 2009). The induction of the RpoS regulons is linked with a wide-range of the stress conditions (Battesti et al., 2011). This sigma factor is known for the regulating cell shape, biofilm formation, DNA repair, virulence factor genes and stress responses (Chiang and Schellhorn, 2010). This conveys the role of RpoS sigma factor in acquiring AMR as it impacts the virulence potential of pathogens in a stressed environment. For example, Gutierrez et al. (2013) observed increased mutagenesis in antibiotics-treated bacterial cells because of the induced RpoS regulon. Hence, the RpoS greatly influences the AMR pattern of the pathogens due to its direct link with mutation rate, which may significantly alter the process of resistance development (Denamur and Matic, 2006). When antibiotic-mediated stress leads to the DNA damage, it gets repaired via the SOS pathway with the help of alternative polymerase. However, alternative polymerase sometimes may turn error prone and therefore lead to mutation(s). The RpoS pathway also enhances the mutation rate in a similar manner.

Since antibiotics evoke the bacterial stress responses that in turn results in acquiring resistance against them; the concentration of the antibiotics also plays a vital role in resistance development. The event of mutation improves the possibility of attaining AMR; therefore, the knowledge regarding the mutant selection window is crucial to study the development/ progression of AMR. The mutant selection window represents the range of antibiotic concentration from the minimal dose required to inhibit the growth of wild type bacterial strain to that required to inhibit the growth of least susceptible strain (Drlica, 2003). Dimension of this window is a characteristic feature of the particular antibiotic-pathogen pair, and the antibiotics can be ranked on the basis of antimicrobial pharmacokinetics. Application of antibiotics within the range of mutant selection window selectively enriches the population of resistant mutants whereas antibiotic concentration exceeding the window restrict selective enrichment (Drlica, 2003).

The resistance gene(s) localized on bacterial mobile genetic elements play key roles in acquiring and spread of AMR via horizontal gene transfer (HGT) (Normark and Normark, 2002). These genetic elements include the self-replicating R-genes harboring plasmids, integrons, transposons and resistant islands frequently transporting the resistant genes into the susceptible strains via HGT mechanisms such as transformation, conjugation or transduction (Li et al., 2019). These mobile elements further utilize site specific recombinases/integrases for their integration in the host genome (Davison, 1999; Rowe-Magnus and Mazel, 2002). It is well-evident that the acquisition of AMR is achieved via mutation and/or HGT (Davies, 1995). Besides, SOS responses are also known to induce the genes involved in HGT. Interestingly, SOS response and the HGT is facilitated at sub-inhibitory antibiotic concentrations of the antibiotics (Beaber et al., 2004; Ubeda et al., 2005; Guerin et al., 2009), stimulating the transfer of integrons, transposons, insertion sequences, pathogenicity islands and integrating conjugative elements (Aminov, 2011).

Apart from these molecular determinants of AMR, the persistence of antibiotics in the environment has also emerged as a key driving factor for AMR and its spread. Besides, studies have confirmed that the enrichment and the maintenance of the AMR are possible in pre-existing resistant pathogen even at very low antibiotic levels (Gullberg et al., 2011; Holmes et al., 2016). The occurrence of antibiotics in soil was reported to escalate the HGT and causes the diversity in mobile genetic elements (Heuer et al., 2009). The authors observed a plasmid-mediated interspecies gene transfer and subsequent spread of AMR gene (sulfonamide resistance gene; *sul1/2*) from *Acinetobacter* spp. to soil bacterial communities (Heuer et al., 2009). Similarly, the effluent from the pharmaceutical industries has also been reported to pose a risk of acquiring and spread of AMR in the environment. A recent study by González-Plaza et al. (2019) revealed the effluents of antibiotic-manufacturing sites as hot-spots for release as well as spreading of AMR-genes and mobile genetic elements in the receiving aquatic environments. Further, the spread of AMR apart from the nosocomial sources

is also well marked due to the presence/usage of antibiotics in agriculture and allied practices (Heuer et al., 2011), waste-water treatment (Karkman et al., 2018) as well as in food processing (Mathur and Singh, 2005).

3. Combating antimicrobial resistant pathogens with phytochemicals

Since ancient times, healing properties of the medicinal plants (and products thereof) are well-acknowledged in combating the human infections. Recent investigations have enlightened the knowledge regarding the antimicrobial activities of PNPs (or plant secondary metabolites, PSMs) against the infectious human pathogens (Barbieri et al., 2017). Phyto-antimicrobials are gaining momentum attributed to their unique structures, broad mechanism of action and safety profiles. Employing phytochemicals against diverse pathogenic microbial strains either for direct bactericidal or as resistance-reversal (re-sensitizing) agents, present an upcoming and prospective strategy for combating the threatening AMR (Khameneh et al., 2019).

3.1. Direct bactericidal activities

Over the previous decade, the phytochemicals are being tested for their antibacterial activity against most prominent drug-resistant bacterial strains (reviewed by Shin et al., 2018). Investigations have been carried out to show the antimicrobial potentials of phyto-chemicals/ extracts showing either direct bactericidal activities or synergistic effects when used with other synthetic antibiotics of non-plant origin.

Several classes of phytochemicals have been screened for identifying the potential bactericidal compounds. These include alkaloids, flavonoids, glycoalkaloids, lactones, phenols, saponins, sesquiterpenes, tannins and terpenoids (Lewis and Ausubel, 2006). Most of these phytochemicals are produced as a plant defense response against the biotic invasions or environmental challenges (Verma and Shukla, 2015). There are numerous reports on the antibacterial activities of phytochemicals which kills the pathogen via a variety of mechanisms. Some recognized modes of action include blockage of bacterial DNA synthesis, inhibition of ATPase activity and membrane disruption (Suarez et al., 2005; Simões et al., 2009). However, the detailed knowledge regarding the exact mode of action of many phytochemicals is yet to be established and needs more investigations.

The phytochemicals with distinct antimicrobial mechanisms against diverse pathogens includes Gallic acid from berry (active against *Salmonella* spp. and results in permeabilization of outer membrane) (Nohynek et al., 2006), Carvacrol (against *Bacillus cereus* by altering membrane permeability) (Ultee et al., 1999), Monoterpenes- Menthol, Linalyl acetate, Thymol (inhibit *Staphylococcus aureus* and *E. coli* by intracellular material leakage and change in membrane permeability) (Trombetta et al., 2005), Berberine (restricts *P. aeruginosa* via inhibiting the efflux pump *MexAB-OprM*) (Aghayan et al., 2017), Quercetin (inhibits *E.coli* DNA gyrase and ATPase activity) (Plaper et al., 2003), epigallocatechin gallate (inhibits the trans-2-enoyl-ACP reductase (FabI) and the β -ketoacyl-ACP reductase (FabG) in bacterial type II fatty-acid synthase system) (Zhang and Rock, 2004). Further, plant essential oils have been reported to exhibit effective antimicrobial potencies via targeting drug-resistance, pathogenic determinants, quorum sensing, microbial biofilms and R-plasmids (Yu et al., 2020). A summarized list of important phytochemicals, their source and antimicrobial potencies is given in Table 1.

3.2. As drug-adjuvants for re-sensitizing antimicrobial resistant strains

Many phytochemicals do not hold the substantial antibacterial potential on their own; however, their application along with other drugs may considerably augment the antibiotic potential of the drug against which the pathogen was resilient. This act is either referred to as the

Table 1

A summarized list of selected phytochemicals, their source and bactericidal and drug-resistance reversal activities.

Targeted mechanism	Source	Phyto-molecule (class)	Organism (plasmid/Efflux pump)	Reference
A. Direct Bactericidal Action of phytochemicals				
Biofilm formation	EOs of lemongrass, lime, ginger, lavender <i>Origanum vulgare</i>	Geraniol (<i>acyclic monoterpene alcohol</i>)	<i>S. epidermidis</i> RP62A	Kannappan et al. (2017)
		Oregano oil	MDR <i>A. baumannii</i> , MDR <i>P. aeruginosa</i> , Methicillin resistant <i>S. aureus</i>	Lu et al. (2018)
Membrane integrity and/or permeability	Components of EOs <i>Populus balsamifera</i> Components of EOs	Carvacrol (<i>phenolic monoterpene</i>)	<i>S. aureus</i> , <i>S. epidermidis</i>	Nostro et al. (2009)
		Balsacone C (<i>dihydrochalcone</i>)	Methicillin resistant <i>S. aureus</i>	Côté et al. (2019)
Quorum sensing	Menthol, thymol, and linalyl acetate (<i>monoterpenes</i>)	Gallic acid (<i>phenolic acid</i>)	<i>Salmonella</i> spp.	Nohynek et al. (2006)
	Berry Components of EOs	Carvacrol	<i>Bacillus cereus</i>	Ultee et al. (1999)
–	<i>Allium sativum</i> <i>Curcuma longa</i>	Ajoene (<i>organosulfur compound</i>)	<i>P. aeruginosa</i> PAO1	Jakobsen et al. (2012)
		Curcumin (<i>polyphenol</i>)	Methicillin resistant <i>S. aureus</i>	Teow et al. (2016)
–	<i>Populus balsamifera</i> <i>Allium sativum</i>	Balsacone A, B, C (<i>dihydrochalcone</i>)	<i>E. coli</i> , <i>S. aureus</i>	Lavoie et al. (2013)
		Allicin (<i>organosulfur compound</i>)	Methicillin resistant <i>S. aureus</i>	Cutler and Wilson (2004)
–	<i>Bersama abyssinica</i>	Picolinyl hydrazide	MDR uropathogenic <i>S. aureus</i>	Ameyia et al. (2019)
B. Drug-resistance reversal or re-sensitizing of antibiotics by phytochemicals as adjuvants				
Enzyme modification	<i>Taraxacum</i> spp.	Taraxasterol (<i>Triterpene</i>)	<i>E. coli</i> K12-29425	Yang and Zhang et al., (2020)
		1,4-naphthalenedione (<i>Quinone</i>)	Amoxicillin-resistant <i>S. aureus</i>	Vinod et al. (2010)
R-genes harboring plasmids	<i>Holoptelea integrifolia</i> <i>Rumex vesicarius</i>	clavulanic acid (<i>Clavam</i>)	Amoxicillin-resistant <i>P. aeruginosa</i>	Al-Sahli and Abdulkhair (2011)
		2-hydroxy-1,4-naphthoquinone (<i>Quinone</i>)	<i>A. baumannii</i> A24 (pUPI281) and <i>E. coli</i> (pRK2013)	Patwardhan et al. (2018)
–	<i>Dioscorea bulbifera</i>	8-epidiosbulbin E acetate (<i>Norditerpene</i>)	<i>E. coli</i> (pAR1813, RP4), <i>P. aeruginosa</i> (RMS163, RIP64), <i>E. faecalis</i> (pAR1812), <i>S. sonnei</i> (pAR1815), <i>B. subtilis</i> (pUB110)	Shriram et al. (2008)
	<i>Alpinia galanga</i>	1'-acetoxychavicol acetate (<i>Phenylpropanoid</i>)	<i>E. coli</i> (pAR1813, RP4), <i>S. Typhi</i> (pAR1814), <i>P. aeruginosa</i> (pAR1816), <i>E. faecalis</i> (pAR1812), <i>B. cereus</i> (pAR1817)	Latha et al. (2009)
Drug efflux pumps	<i>Berberis vulgaris</i>	Berberine (<i>Alkaloid</i>)	MDR <i>P. aeruginosa</i> (clinical isolates) (MexAB-OprM)	Aghayan et al. (2017)
		Conessine (<i>Alkaloid</i>)	<i>P. aeruginosa</i> (MexAB-OprM)	Siriyong et al. (2017)
–	<i>Holarrhena antidysenterica</i> <i>Eucalyptus tereticornis</i>	Ursolic acid and derivatives (<i>Triterpene</i>)	MDR <i>E. coli</i> (KG4) (AcrA/B, MacB, TolC and YojI)	Dwivedi et al. (2015)
	<i>Terminalia chebula</i>	Gallotannin (<i>Tannin</i>)	MDR uropathogenic <i>E. coli</i>	Bag and Chattopadhyay (2014)
–	<i>Ipomoea muricata</i> <i>Hypericum olympicum</i>	Lysergol (<i>Clavine alkaloid</i>)	<i>E. coli</i> (ABC pump YojI)	Maurya et al. (2013)
		Olympicin A (<i>Acylphloroglucinol</i>)	<i>S. aureus</i> 1199B (NorA)	Shiu et al. (2013)

“–”: Not mentioned.

drug resistance reversal or re-sensitizing the resistant strains (Upadhyay et al., 2014). Such phytochemicals are usually applied at lowest possible dose, causing the highest possible drug resistance reversal effect. This effectively leads to the cost-effectiveness and shortening of the treatment course (Cottarel and Wierzbowski, 2007). Phytochemicals target major drug-resistance mechanisms in bacteria like active drug-efflux pumps (Dwivedi et al., 2014) and R-genes harboring plasmids (Shriram et al., 2008), among others. Phyto-therapeutics including pure compounds and plant extracts has been explored as potent bacterial efflux pumps inhibitors (EPIs) (Shriram et al., 2018). Some key examples of phytochemicals with EPI potential consist of resveratrol, pinene, catechol, and gingerol (Prasch and Bucar, 2015). Gallotannin, an effective EPI, was isolated from hydro-alcoholic extract and demonstrated potent antimicrobial activity against MDR uropathogenic *E. coli* (Bag and Chattopadhyay, 2014). The phytochemical inhibited the ethidium bromide (EtBr) pump and showed a decline in minimum inhibitory concentration (MIC) of antibiotics. In another study, lysergol, a clavine alkaloid from *Ipomoea muricata* was tested for antimicrobial (EP inhibition) activity against *E. coli* strains (MTCC1652 and KG4). The phytochemical, lysergol and its derivative 17-O-3'',4'',5''-trimethoxybenzoyllysergol (Maurya et al., 2013) demonstrated strong EP

inhibition (more than reserpine standard), also showing significant inhibition of ABC pump YojI in *E. coli* (Maurya et al., 2013), please refer Table 1.

In combinational therapies, the essential oils with the existing antibiotics exhibit excellent prospects in combating AMR in pathogens (Yu et al., 2020). Considering that most compounds produced in response to pathogen infection are not antimicrobials, an effective synergistic action results when the existing antibiotics are combined with a resistance-antagonizer (Wagner and Ulrich-Merzenich, 2009). Studies have suggested that these phytochemicals have the potential to restore the efficacy of antibiotics in combinational therapies (Barbieri et al., 2017). The ultimate result of these interactions is loss of dynamic antibiotic gaining mechanism to the bacteria; and turning the resistant strains into sensitive. The examples of phytochemicals mediated reversal of the antibiotic resistance are quoted in Table 1 with a schematic in Fig. 1.

4. Limitations of microbial hosts in producing phytochemicals

Considering the limited distribution and/or availability of phytochemicals in natural systems, heterologous production of phytochemicals in suitable hosts presents an emerging and effective approach for

maintaining their commercial-scale supply. However, challenges in structural complexity and metabolic pathways restrict the pathway-reconstitution in heterologous systems (Kotopka et al., 2018). In initial years, bio-based production of phytochemicals was tried using microbial systems, for example *E. coli* and yeast as an alternative supply platform and screening systems for targeted metabolite production. Advances in metabolic engineering have enabled simple carbon source mediated production of diverse small molecules (Zhang et al., 2018), with computational biology and SynBio strategies aiding in PNP-based drug discovery studies. Several metabolic engineering strategies on host level (regulation of metabolic network, genome manipulations and re-direction of carbon flux) and enzyme level (assembly of metabolic pathway, screening enzymes) were established for phytochemical production in microbial hosts (Zhang et al., 2018). However, major challenges that limit the universal application of advanced microbial production of phytochemicals include: 1) incomplete elucidation of biosynthetic pathways of phytochemicals, therefore transfer of pathways from native host to heterologous system is limited, 2) low turnover of metabolic flux in microbial hosts which could be addressed through engineering of host organism for achieving higher expression levels of enzymes, and 3) less activity of some plant enzymes in microbial hosts (Li et al., 2018a, 2018b). The key issues in heterologous pathway reconstitution are metabolic pathway functioning and enzyme optimization in microbial hosts. The emerging trends in SynBio and metabolic engineering approaches aims to promote heterologous expression and functions of metabolic pathways to address the associated limitations (Becker and Wittmann, 2016). Although *E. coli* is the widely used microbial hosts for PNP production, the lack of post-translational modifications and intracellular membranes restrict its use as chassis organism for natural products biosynthesis. Similarly, despite several advantages, engineering *S. cerevisiae* may be a challenging task, owing to non-functional nature and negligible activity of some of its proteins and low yields among others (Dominguez et al., 1998). Moreover, the poor function of yeast spliceosome hampers the introns removal from protein-coding sequences in the heterologous system, leading to less precursor supply and low-titer of the desired metabolite (Moses et al., 2017). In this direction, non-canonical bacterial host, *Bacillus subtilis* with rapid multiplication rate was used for production of proteins and was considered as an ideal choice for terpenoid biosynthesis (Julsing et al., 2007). However, it shows plasmid instability and its partial characterization pose challenges in genetic manipulations.

5. Plant synthetic biology for production of phyto-antimicrobials

Plant SynBio is emerging as a reliable and sustainable platform for producing desired plant metabolites at industrial scale employing plant chassis. The considerable interest in plants as “cell factories” for bioactive PNP production is gaining momentum (Buyel, 2018). According to European Medicinal Agency and Food and Drug Administration, a quarter of currently approved drugs are of plant-origin, owing to their distinct properties including structural uniqueness, broad-spectrum activity, low toxicity etc. making them ideal leads in drug discovery programs (Ajayi et al., 2019). Plant cell cultures have emerged as sound biotechnological platforms for large-scale production of potent PNPs (Marchev et al., 2020; Shaikh et al., 2020) and their large-scale exploration using bioreactor technologies may ensure a sustainable supply of phyto-antimicrobials.

Antimicrobial peptides (AMPs) are integral components of innate immunity in all living organisms and constitute defense molecules against bacteria, fungi, viruses, and other parasites (Holaskova et al., 2015). AMPs are gaining momentum as effective therapeutic agents. Heterologous expression of AMPs in plant systems highlights a potential method for their large-scale and cost-effective production for agricultural or clinical applications (Holaskova et al., 2015). The AMPs can be expressed at a stable level and with sufficient yield in plants, for

example, AMP magainin analogs were stably expressed in tobacco and the transgenic lines showed increased resistance to bacterial pathogens (Li et al., 2001). A synthetic peptide MSI-99 was expressed in a gene-cassette including 16S rRNA promoter, the *aadA* gene, photosynthetic binding protein (psbA) and the MSI-99 gene with a chloroplast preferred (GGAGG) ribosome-binding site (DeGray et al., 2001). Some other successful studies on stable AMP expression in plant systems include transgenic *Nicotiana tabacum* plant expressing Lactoferricin B showed improved resistance to fungal and bacterial diseases (Fukuta et al., 2012), while transgenic *A. thaliana* expressing Thanatin acquired fungal and bacterial resistance (Wu et al., 2013). Plant systems offer better platforms for AMP expression, as bacteria and yeasts have limitations in terms of AMP instability, difficulties in carrying out post-translational modifications besides the possibility that AMPs may arrest the host growth (Chahardoli et al., 2018). Moreover, transgenic plant systems for AMP expression are easy to handle, lesser prone to contamination, and show greater stability besides their ability to produce biologically active proteins owing to correct folding and post-translational modifications (Wani and Sah, 2015). However, problems may arise while incorporating the non-canonical amino acids, for example, adding D-amino acids to the backbone and creating cyclic bridges to obtain modified AMPs.

Some significant reviews have discussed the prospects of phytochemicals and their antimicrobial potentials, a few key ones include employing phytochemicals to counteract biofilm formation and resistance (Borges et al., 2016), antimicrobial potential of essential oils to counter AMR (Yu et al., 2020), phytochemicals as antimicrobials and resistance-modifying agents (Abreu et al., 2012), and phytotherapeutics as inhibitors of bacterial drug efflux pumps (Shriram et al., 2018). Initially microbial systems were extensively employed for PNP production via SynBio, however, plant chassis offers prospective yet less-explored approach for phytochemical production. Fig. 2 depicts major approaches for the use of plants as chassis for production of potent phytochemicals against antibiotic resistant microbes.

5.1. Plant hosts as sustainable and engineerable platforms for production of antimicrobial agents

SynBio strategies have focused on delineating the biological processes to chassis organisms for engineering applications. Contemporary targeted metabolic engineering approaches aim to promote plant utility beyond food resource to renewable energy source, industrial metabolites and therapeutics (Shih, 2018) and plant systems look as a key sustainable platform for large-scale and targeted PNP production. Antimicrobial agents from synthetic sources were traditionally approved for clinical applications, however, natural products from plants, animals and microbial sources are now gaining an unprecedented attention from the scientific community. The efficacy of PNPs in combating AMR highlights an effective approach to curb microbial diseases (Rossiter et al., 2017). Besides, studies have shown that co-administration of phyto-antimicrobials with existing antibiotics promotes synergistic action and lowers minimum inhibitory concentration (MIC) of the antibiotics (Khameneh et al., 2019).

SynBio has opened new avenues in discovery of antimicrobials, through manipulation of small peptides and molecules and non-traditional antimicrobials. For instance, antimicrobial peptides (AMPs) have increasingly been recognized as potential alternatives to conventional antibiotics, attributed to their broad-spectrum function, specificity of targets, distinct mechanism and low cytotoxicity (Holaskova et al., 2015; da Cunha et al., 2017). However, AMP production in plants need to take care of some structural aspects, namely small size, presence of amino acids (protease cleavage) and target sequence, and inclusion of retention signals for an efficient heterologous expression (Badosa et al., 2013). Genetic engineering approaches were adopted for production of novel AMPs in plants including *Nicotiana tabacum*, *Lycopersicon esculentum*, *Solanum tuberosum*, and *Musa paradisiaca*, beside others. Some

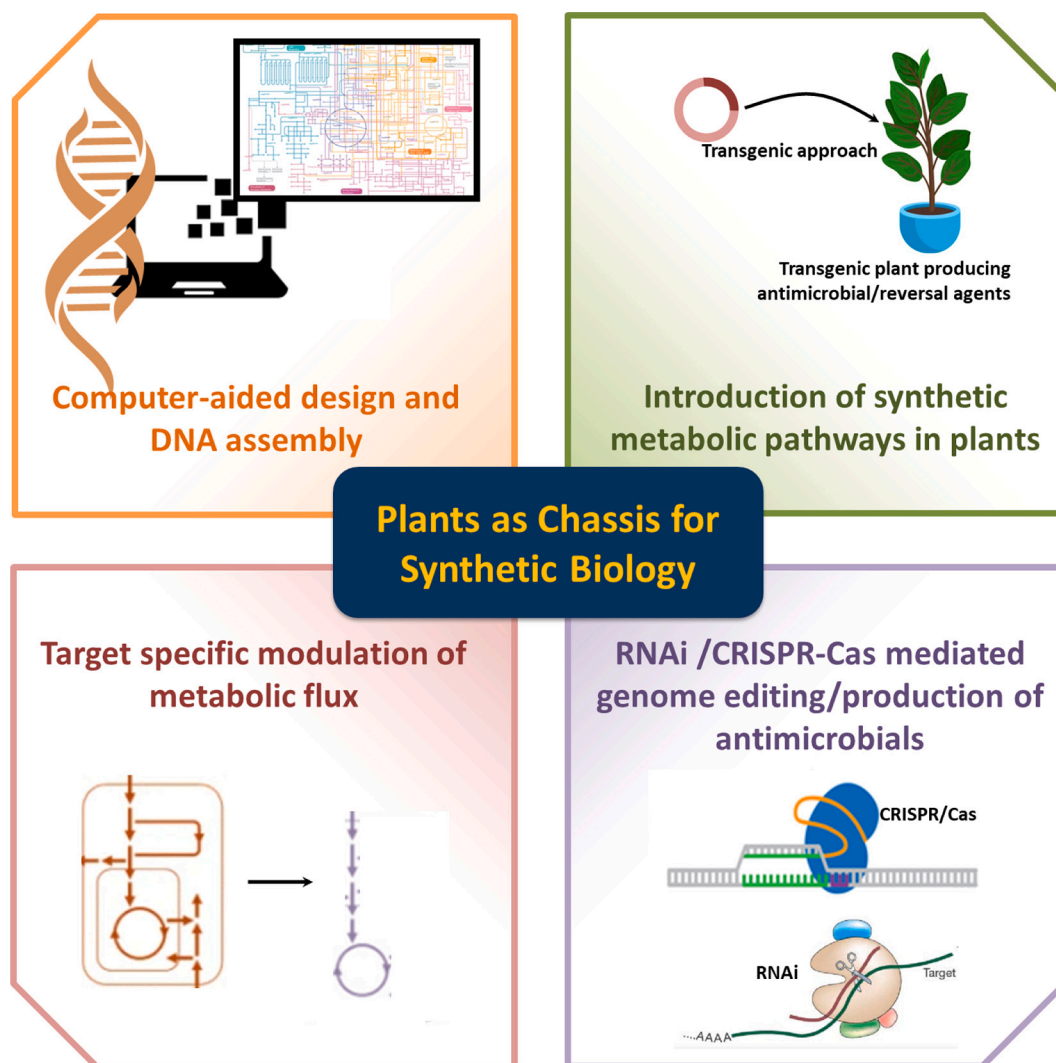


Fig. 2. Approaches depicting the use of plants as chassis for production of potent phytochemicals against antibiotic resistant microbes.

key examples of AMPs produced via plant metabolic engineering approaches are highlighted in Table 3. SynBio strategies have made significant contributions in plant chassis for AMP production, for instance, employing RNAi in defense mechanism (Obbard et al., 2009), molecular farming for production of phyto-antimicrobials (e.g. peptide LL-37 in *Hordeum vulgare*) (Holásková et al., 2018) and genetic transformation (e.g. SN1 AMP in *T. aestivum*) for tolerance to pathogen (Rong et al., 2013), refer to Table 3. Plant systems therefore represent a renewable source of high-value therapeutics and form an ideal chassis platform for antimicrobial inventories, subject to pathway reconstitution and heterologous expression via SynBio.

5.2. Synthetic biology to develop plant chassis

Plants as “cell factories” highlight advantages over their microbial counterparts, making them ideal platforms for manipulations via SynBio (Lau et al., 2014). With advances in high-throughput technologies, molecular and genomics studies in non-model plants have revolutionized, gene organization and expression and cataloging of specialized metabolites have become more evident in evolutionary analysis (Unamba et al., 2015). Whole genome sequencing and an ever-increasing dataset on plant systems have facilitated the cataloging of species-specific metabolites and their bio-prospection in the future. Besides precise cataloging, an expansion of SynBio tools and engineering principles however is required to explore the unique targets and their

manipulations. Identification of gene families and metabolic pathways involved in biosynthesis of PNPs highlight a prospective platform to reconstruct plant pathways in heterologous systems. Manipulation of *in vitro* plant systems for enhanced production may be targeted by the integral characteristics of plant cells to synthesize phytochemicals. While, the microbes have an edge as “production platforms” with respect to well-developed genetic tools and culture methods for production of a specific metabolite or inclusion of a plant metabolic pathway, *in vitro* plant systems offers a sustainable approach to produce PNPs (Maeda, 2019). Several challenges exist in microbial chassis for PNP production as discussed earlier in Section 4, for instance, poor activity of plant enzymes and low metabolic flux of intricate plant metabolic pathways in microbial systems (Li et al., 2018a, 2018b). Further, the intracellular environment of microbes differs from those of native plant hosts, and heterologous expression of PNPs may face additional challenges of lower levels of expression, incorrect folding, and reduced catalytic efficiency than plant systems. For example, improving the heterologous activity of plant P450s was found challenging for microbial based production of complex phytochemicals (Li et al., 2018a, 2018b). On the other hand, plant *in vitro* systems are emerging as sound and scalable platforms for metabolite and recombinant protein production at low manufacturing costs (Giddings, 2001; Maeda, 2019).

In recent times, there has been growing interest and initiatives toward developing plant chassis to enhance *in planta* production of value-added metabolites, particularly of medicinal importance. Although

plants as natural producers of pharmacological compounds are extensively explored, biosafety concerns are often raised on administration of crude-extracts (Negi, 2012). Additionally, metabolites possessing pharmacological properties (in low concentrations) endangers plant population. A case study on large-scale production of Taxol from Pacific yew raised concerns of threatening this plant (Hawkins, 2008). However, the engineering of the biosynthetic pathway necessitates knowledge of the metabolic flux, core pathways, metabolic constraints, and the effect of metabolic channeling on the system (Barone et al., 2019). Microbial systems as chassis platform projects additional complexities: the pathway interaction with the endogenous host metabolism leading to formation of undesirable intermediates and end-products (Farré et al., 2014). Recent advances in technological interventions are opening new avenues in natural product engineering. Strategies in plant chassis include increasing the expression of upstream pathway enzymes for efficient precursor supply, increasing expression of first enzyme towards a targeted metabolite production and enhancing function of rate-limiting enzymes (which leads to accumulation of inhibitory intermediates) among other strategies of silencing linked pathways and limiting catabolism of metabolites leading to a targeted approach for PNP production (Farré et al., 2014).

For introduction of transgenes in plants, both direct and indirect delivery methods have been standardized and their selection may depend on the basis of plant host and the expression vector in consideration. Two of the most often used technologies representing direct and indirect delivery methods are biolistic gene-gun and *Agrobacterium*-mediated gene transfer, respectively. Biolistic gene-gun based delivery system offers a vector-less approach effectively used for transferring/overexpressing gene(s) in chloroplast or nuclear genomes of virtually any plant species. The plant cells are bombarded with high-speed particles (coated with target nucleic acid) which penetrate the plant cells and transgene incorporation takes place (Lacroix and Citovsky, 2020). Successful gene transfer via microprojectile bombardment approach is reported in several medicinal plants including *Cuminum cyminum* (Singh et al., 2010), *Centella asiatica* (Lai et al., 2011), *Catharanthus roseus* (Guirimand et al., 2009), *Jatropha curcas* (Joshi et al., 2011), and *Scoparia dulcis* (Srinivas et al., 2016). However, historically, one of the most frequently used methods of transgene delivery includes *Agrobacterium*-mediated plant transformation (Chen and Lai, 2015). This is mainly because of the advantages this method offers in terms of low copy number of transgenes and better control of genome integration. Besides, a high expression level can be achieved owing to the co-evolution of *A. tumefaciens* with its host plants which favors the transgene integration into the transcriptionally active genomes resulting in high expression levels (Klee et al., 1987; Chen and Lai, 2015). Agro-infiltration method addresses the problem of random transgene insertion into the genomes, by facilitating the active delivery of transgene-carrying agrobacteria into the leaf tissue, improving the efficiency of the transformation (Vaghchhipawala et al., 2011).

Viral-based expression systems constitute attractive alternatives for protein overexpression in plants and rely on replicating plant viruses. In this method, the target gene is incorporated into the viral elements, amplified and translation takes place in plant cytosol. Some earlier attempts employing plant viruses for transgene delivery (cauliflower mosaic virus, CaMV) faced limitation of packaging capacity and restricted removal of non-essential viral DNA. Gene substitution vectors constitutes an endogenous viral sequence, including the target gene for heterologous expression. One of the key successes in expression of a foreign gene in plants was achieved by employing CaMV substitution vector (Scholthof et al., 1993). Gene insertion vectors include complete viruses with an open reading frame for the target protein and capable of systemic infection and movement. One of the gene insertion vector was derived from Tobacco Mosaic Virus and comprises of single-stranded RNA genome, however, challenges regarding recombination between duplicated sub-genomic promoters were reported for gene insertion vectors (Haviv et al., 2006), which limits their use. The current research

initiatives are directed towards development of novel viral vectors in terms of improved expression methods and virus selection.

Plant chassis for expression of single genes is usually straightforward, however, multi-gene expression requires a coordinated control and may be a challenging task. The overexpression strategies often employ T7 RNA polymerase and its natural T Φ Class I terminator (Du et al., 2009). The objectives range from coordinated expression of metabolic pathways to gaining insights on gene regulation and reprogramming cellular functions through transcriptional mechanisms (Machens et al., 2017). For well-tuned gene expression, initial studies were based on mono-directional promoters which had limited capacity for co-expression of multi-genes. With the aim to improve the flexibility for optimization and assemblage of metabolic pathways, the bidirectional promoters were engineered for efficient co-expression of multi-genes (Vogl et al., 2018). For example, a library of 168 synthetic bidirectional promoters was generated in *Komagataella phaffii*, for optimization of gene co-expression for metabolic pathways and fast screening of different expression profiles (Vogl et al., 2018). Moreover, employing synthetic transcription factors (synTFs) enables the control of different gene expressions (Lu et al., 2009). Although, synTFs are implemented in transcription regulation but the DNA-binding specificity is generally non-variable, leading to development of synTFs with programmable DNA-binding specificity (Machens et al., 2017). The initiatives on synTFs development started with DNA-binding proteins (based on zinc finger domains) with transcription activator-like effectors (TALEs) and RNA-guided CRISPR-Cas9-based TFs being the recent class with high flexibility (Wiedenheft et al., 2012; Naseri and Koffas, 2020). The flexibility of using synthetic promoters and terminators provides an additional advantage in constructing tailored transcriptional control systems.

Considerable potential for high-level production of pharmaceuticals with less expense is demonstrated by plant systems, however, high cost in downstream extraction and concerns regarding biosafety (e.g. immunogenic plant glycans) have somewhat limited the production of pharmaceuticals on an industrial scale (Yao et al., 2015). Besides, currently used technologies cannot precisely control transgene expression in plants and the difficulties associated with plant transformation. The production of pharmaceuticals might show variations amongst the species/ecotypes with differential expression levels, leading to bottlenecks in accurate quantification of edible vaccine dosages for the patients (Yao et al., 2015). Although plant systems offer reservoirs of the drug/drug-like molecules, low abundance and limited access to the plant species may hamper the optimal use. The science of plant chassis for PNP production being a relatively new field needs careful attention and effective solution to these issues and necessitates further research.

However, despite such bottlenecks, SynBio strategies have proved to be successful to a considerable extent and hold promise for greater exploration (Lu et al., 2013). A comprehensive knowledge of plant primary and secondary metabolism, complexity of plant metabolic networks, a constrained plant metabolism and little space for manipulation of plant genomes contributes to bottlenecks in manipulating *in vitro* plant systems (Maeda, 2019). In addition, the closely linked plant pathways and their direct effect on plant physiology and an exclusive dependence on carbon supply from photosynthesis, restricts plant manipulations. A plausible solution and way forward in plant chassis is to screen plant-specific secondary metabolite production, and a precise understanding of precursor supply from primary metabolism. An example showed that an effective precursor supply (anthranilate and tyrosine) in low concentrations in *Nicotiana benthamiana* stimulated the downstream biosynthetic enzymes and enhanced betalains production (Timoneda et al., 2018). The knowledge about the diversification of primary metabolic pathways during evolution in plants would serve as a platform for reconstruction of a PNP pathway through genetic engineering (Maeda, 2019). *Nicotiana benthamiana* is gaining popularity as an expression platform for plant pathways reconstitution, attributed to its responsiveness for *Agrobacterium*-mediated transformation and

scalability for rapid gene expression (Reed and Osbourn, 2018). Additionally, as it shares common cellular compartmentalization, cofactors and coenzymes with other plants, transferring of pathways from other plants does not require extensive optimizations, and thus seems an ideal platform to be explored as a potent chassis for SynBio. These properties facilitate expression of biosynthetic libraries (in combination) without the requirement of multi-gene constructs (Reed and Osbourn, 2018). The presence of metabolic precursors in *N. benthamiana* aids in function of heterologous expressed enzymes and may be subjected to leaf infiltration (Miettinen et al., 2014) used for determining enzyme functions. Reconstitution of different PNP- pathways was successful in *N. benthamiana*, including betalains (Polturak et al., 2016), cyanogens (Rajniak et al., 2015), alkaloids (Miettinen et al., 2014), glucosinolates (Crocchi et al., 2016), highlighting the potential of this plant system for production of high-value metabolites. The need for affordable and consistent supply of high-value metabolites and the associated challenges in obtaining these metabolites from natural sources have prompted researchers to engineer plant systems. Transient technologies (e.g. agro-infiltration) have been used for engineering PNP pathways to produce purified triterpenes in good quantities (gram-scale) (Reed et al., 2017; Reed and Osbourn, 2018). A combinational biosynthesis of triterpenes in scalable quantities (without multi-gene constructs requirement) in *N. benthamiana* and its purification and bioassays provides new

routes to biosynthesize inaccessible yet high-value metabolites via SynBio. The transient expression systems like *N. benthamiana* can be used to test/design 'novel' phytochemicals that are not found in nature, which then can be tested for their antimicrobial/re-sensitizing activities.

A key study has focused on the metabolic engineering of *Mentha piperita* (Peppermint) by transferring the homologous sense version of 1-deoxy-d-xylulose-5-phosphate reductoisomerase cDNA and 1-deoxy-d-xylulose-5-phosphate reductoisomerase cDNA (both regulated by CaMV 35S promoter). The transgenic plants showed normal growth and morphology, and accumulated 50% more essential oils (Mahmoud and Croteau, 2001). In an attempt, carbon flow was re-directed from plastidic or cytosolic isopentenyl diphosphate through overexpression and it increased the biosynthesis of patchoulol and amorpho-4,11-diene (sesquiterpenes) over 1000-folds (Wu et al., 2006). Another attempt towards metabolic engineering of synthetic multigene operon in tomato plastids and transgenic tobacco uncovered a light-dependent regulatory association between the photosynthesis pigment biosynthesis pathways and tocopherol metabolism (Lu et al., 2013). In a noteworthy study, three crops namely *Solanum tuberosum*, *S. lycopersicum* and *S. melongena* were used for heterologous betalain production (Polturak et al., 2017). Studies on genetic engineering of *Arabidopsis thaliana* for heterologous expression of maize terpene synthase (TPS10) was performed which resulted in enhanced production of sesquiterpenes *in planta* (identical to

Table 2

A summarized list of examples using plant chassis for synthetic biology (SynBio) strategies for altered secondary metabolite production.

S. No.	Secondary metabolite class (Example)	Plant system	Synthetic biology strategy	Biological significance	Reference
1.	Alkaloid (Strictosidine)	<i>Nicotiana benthamiana</i>	Pathway reconstitution in heterologous plant	Elucidation of Strictosidine biosynthetic pathway	Miettinen et al. (2014)
2.	Alkaloid (Camptothecin)	<i>Camptotheca acuminata</i>	Overexpression of ORCA3 gene	Enhanced Camptothecin production	Ni et al. (2011)
3.	Diterpene (Taxadiene)	<i>Solanum lycopersicum</i>	Redirection of carotenoid metabolism	Metabolic engineering for Taxadiene production	Kovacs et al. (2007)
4.	Diterpene (Taxadiene)	<i>Panax ginseng</i>	Genetic engineering of <i>P. ginseng</i> roots with a taxadiene synthase gene	Enhanced production of Taxadiene	Cha et al. (2012)
5.	Diterpene (Taxadiene)	<i>N. tabacum</i>	Virus-induced gene silencing of phytoene synthase (PSY) enzyme	Enhanced production of Taxadiene	Hasan et al. (2014)
6.	Flavonoid (Luteolin, Apigenin)	<i>Codonopsis lanceolata</i>	Overexpression the γ -tocopherol methyl transferase (γ -tmt) gene	Enhanced production of phenolic compounds and antimicrobial activities	Ghimire et al. (2017)
7.	Flavonoid, Apocarotenoid	<i>L. esculentum</i>	Gene silencing the DET1 regulatory gene	Enhanced Flavonoid and Apocarotenoid content in tomato fruit	Davuluri et al. (2005)
8.	Flavonoids	<i>N. benthamiana</i>	Gateway®-based high-throughput viral vector cloning system	Production of flavonoids	Zhang et al. (2013)
9.	Glucosinolates	<i>N. tabacum</i>	Pathway reconstitution and heterologous expression of three pathway steps	Genetic engineering of glucosinolates into non-cruciferous plants	Geu-Flores et al. (2009)
10.	Isoflavones	Legumes	Overexpression of isoflavone synthase	Production of phytoalexins	Jung et al. (2000)
11.	Phenolic acid (Rosmarinic acid)	<i>S. miltiorrhiza</i>	CRISPR-Cas9-mediated efficient targeted mutagenesis of rosmarinic acid synthase (RAS)	Regulation of phenolic acid pathway and enhanced metabolite production	Zhou et al. (2018)
12.	Phenolic acid (Caffeic acid, Ferulic acid, p-coumaric acid, Chlorogenic acid)	<i>Leonurus sibiricus</i>	Overexpression of the AtPAP1 transcriptional factor in root culture	Antimicrobial and cytotoxic activity	Sitarek et al. (2018)
13.	Sesquiterpenes	<i>Arabidopsis thaliana</i>	Genetic engineering for heterologous expression of maize terpene synthase TPS10	Enhanced sesquiterpenes production for indirect plant defense against herbivore attack	Schnee et al. (2006)
14.	Sesquiterpene (α/β -santalene), Sesquiterpene alcohol (Patchoulol)	<i>Physcomitrella patens</i>	Metabolic engineering of the moss	Heterologous production of Patchoulol and α/β -santalene	Simonsen et al. (2009)
15.	Sesquiterpene ((+)-valencene)	<i>N. benthamiana</i>	RNAi-gene silencing of endogenous squalene synthase (SQS) and the 5-epi-aristolochene synthase (EAS)	Heterologous production of the sesquiterpene (+)-valencene	Zhan et al. (2014)
16.	Sesquiterpene lactone (Artemisinin)	<i>Nicotiana tabacum</i>	Targeted expression of amorphadiene synthase (ADS) in mitochondria	Enhanced Artemisinin production	Cankar et al. (2015)
17.	Steviol glycosides	<i>Stevia</i> spp.	Overexpression of kaurenoic acid hydroxylase (SrKAH) and 1-deoxy-d-xylulose-5-phosphate synthase 1 gene	Increased production of steviol glycosides	Farhi et al. (2011)
18.	Stilbenoid polyphenol (t-resveratrol)	<i>Silybum marianum</i>	Genetic transformation of stilbene synthase gene in plant suspension cultures	Increased accumulation of t-resveratrol	Zheng et al. (2019)
19.	Steroidal lactones (Withanolides)	<i>Withania somnifera</i>	RNAi-mediated gene silencing of cycloartenol synthase (CAS) gene	Enhanced withanolide biosynthesis	Hidalgo et al. (2017)
					Mishra et al. (2016)

maize sesquiterpenes) and improved plant tolerance to herbivore attack (Schnee et al., 2006). Farhi et al. (2011) altered the expression of amoradiene synthase gene in mitochondria of tobacco for enhanced production of artemisinin. Similarly, overexpression of the stilbene synthase gene in suspension cultures of *Silybum marianum* resulted in increased accumulation of t-resveratrol (Hidalgo et al., 2017). Overexpression of the *AtPAP1* TF in root culture of *Leonurus sibiricus* enhanced the caffeic acid, ferulic acid, *p*-coumaric acid and chlorogenic acid production (Sitarek et al., 2018). Further, Cankar et al. (2015) employed an RNA-interference mediated gene silencing mechanism to switch off the endogenous squalene synthase and the 5-epi-aristolochene synthase, increasing valencene production.

Plant metabolic engineering represents an efficient strategy to enhance production of target metabolites. However precise and clear insights of plant metabolic pathways and their coordination and employing novel, prospective avenues in SynBio would aim at a more targeted approach in plant chassis for PNP production. Some important case studies using plant chassis for production of phyto-antimicrobials are presented in Table 2. SynBio played a key role in assembly of complete metabolic pathways, thus providing important insights of plant metabolism (Andrianantoandro et al., 2006). Novel strategies in combinational engineering, selection of target plant library comprising diverse metabolic transgenes would prove useful in checking undesirable reactions in metabolic pathways (Farré et al., 2014). Future initiatives towards plant chassis from single to multi-gene interventions together with high-throughput technologies would revolutionize the gene discovery and novel pathways for combinational testing in plants, improving the predictability of plant chassis for targeted PNP production.

While plant metabolic engineering or potential use of plant chassis for SynBio are emerging revolutionary approaches, however, the public acceptance and use of genetically modified (GM) plants are yet to be completely realized. The development of transgenic plants has tremendously increased the crop production and stress-tolerance, besides improving our understanding of plant biology, however, the cultivation and consumption of GM crops has been a matter of great controversy (Ahmad and Mukhtar, 2017). Further concerns regarding biosafety, environmental concerns and loss of wild varieties because of GM plants

highlight bottlenecks associated with full-potential exploration of transgenic plants (Ahmad and Mukhtar, 2017). Genetic engineering of plants for crop production (food/feed) or production of high-value metabolites faces competition. Though it is important to grow plants with improved traits and nutrient content for feeding millions across the globe, it is equally important to enhance the biotechnological prospects (production of therapeutics) in healthcare. An interesting development towards the use of plant chassis highlights the developing of rice lines that produces antibacterial proteins (present in saliva and human breast milk) by the Ventria Bioscience (Huang et al., 2002). The transgenic rice synthesizing lysozyme and lactoferrin proteins can be used for treating diarrhea in children. While the prospects of developing genetically engineered crops for pharmaceuticals is immense, the concerns with perceived risks to the food supply and environment and strict guidelines for growth containment further adds to limitations (Babcock, 2004).

The natural resources being limited and gradually depleting, alternative strategies like plant cell cultures are needed as an alternative to whole plant systems, for sustainable production of PNPs (Nielsen et al., 2019). Plant cell-lines may be initiated from an explant and propagated to generate multiple tissues and represent an alternative system for producing metabolites regardless of climatic fluctuations and need for agricultural land. The use of alternate production platforms for high-value metabolites will thus facilitate the scalable production of PNPs. In the present scenario, food and chemical industries aim to develop safe and feasible extraction technologies for extracting bioactive compounds. The conventional extraction methods have limitations and the development of newer extraction techniques like instant controlled pressure drop method, supercritical fluids extraction, liquefied gas technology, and mechano-chemical technology are some of the innovative technologies in sustainable extraction of metabolites (Chemat et al., 2020).

5.3. Computer-aided design and target identification

The computational strategies in discovery of antimicrobial agent include simulation and dynamics studies, genomics, quantitative structure-activity relationships, molecular docking, and structure/function prediction methods (Hammami and Fliss, 2010). Recent advances in modeling-based approaches and its application in plant

Table 3

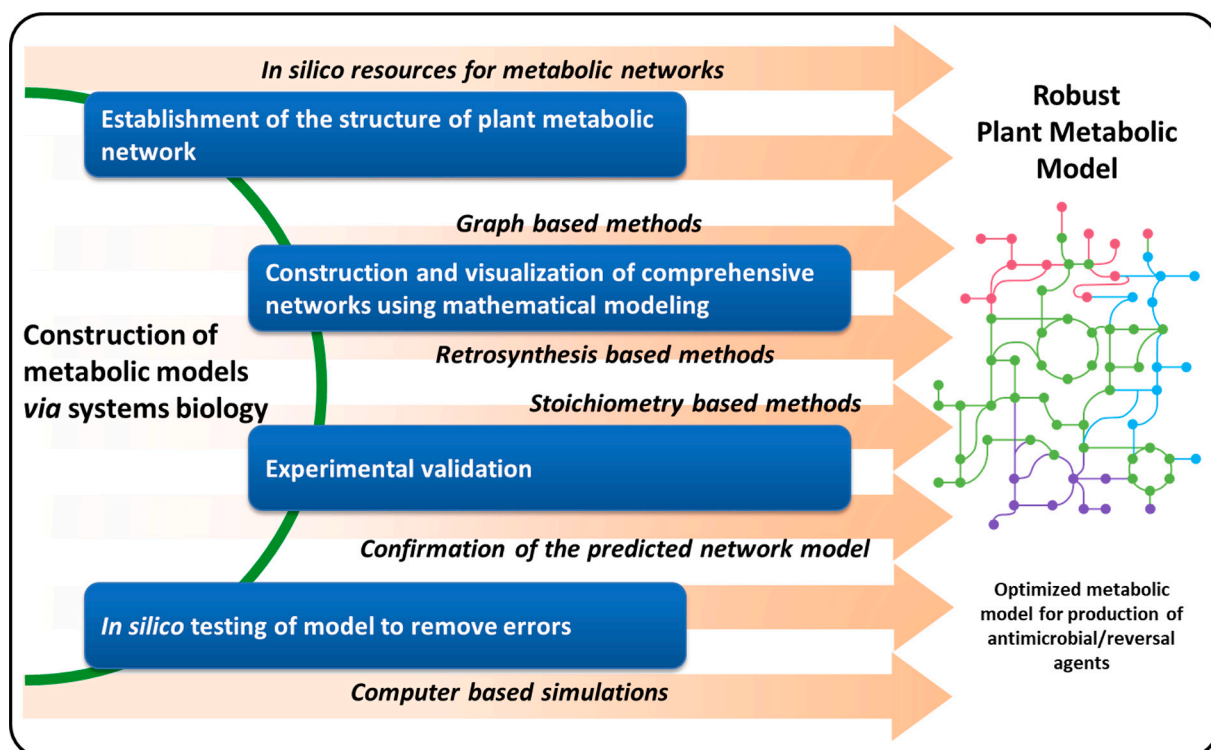
List of major antimicrobial peptides (AMPs) produced in plant systems using metabolic engineering approaches.

S. No.	Antimicrobial Peptide	Plant host	Metabolic engineering strategy	Potential application	Reference
1	Thanatin(S)	<i>Arabidopsis thaliana</i>	Introduction of Cht1 signal peptide (Cht1SP)-thanatin(S)-GFP construct to <i>Arabidopsis</i>	Thanatin(S) demonstrated antibacterial and antifungal activity	Wu et al., 2013
2	Floral defensins	<i>Musa</i> spp.	Overexpression of <i>PhDef1</i> and <i>PhDef2</i>	Resistance against <i>Fusarium oxysporum</i> f. sp. <i>cubense</i> infection and Fusarium wilt disease	Ghag et al. (2012)
3	MsrA2	<i>Poplar</i> spp.	Overexpression of MsrA2	Enhanced resistance to pathogens	Yevtushenko and Misra (2012)
4	Lactostatin	<i>Oryza sativa</i>	Overexpression of A1aB1b containing 29 lactostatins in the endosperm	Enhanced Lactostatin production, anti-hypercholesterolemic drug	Cabanas et al. (2013)
5	Protegrin-1	<i>Nicotiana tabacum</i>	Transfer of RC101 or PG1-GFP construct in chloroplast	Therapeutic agents against bacterial and/or viral infections	Lee et al. (2011b)
6	Retrocyclin-101 rC4V3	<i>N. tabacum</i>	Introduction of recombinant C4V3 into plastome	Potential HIV therapeutic, induce immune response against HIV	Rubio-Infante et al. (2012)
7	SP1-1	<i>N. benthamiana</i>	Designing of AMPs and plant infection by recombinant virus	Antimicrobial activity against plant and human pathogens	Zeitler et al. (2013)
8	Cathelicidin	<i>Brassica rapa</i>	Overexpression of a human cathelicidin antimicrobial peptide (hCAP18/LL-37)	Resistance against fungal and bacterial pathogens	Jung et al. (2012)
9	Snakin-2 (SN2)	<i>Solanum lycopersicum</i>	Over-expression of snakin-2 and Extensin-Like protein genes in transgenic tomato	Increased tolerance to <i>Clavibacter michiganensis</i> subsp. <i>michiganensis</i>	Balaji and Smart, (2012)
10	SN1 antimicrobial peptide	<i>Triticum aestivum</i>	Genetic transformation of SN1 gene	Increased resistance to <i>Gaeumannomyces graminis</i> var. <i>tritici</i>	Rong et al. (2013)
11	Cecropin A	<i>O. sativa</i>	Transformation with cecropin A gene	Resistance to bacterial and fungal pathogens	Bundó et al. (2014)
12	Interferons	<i>Solanum tuberosum</i> <i>O. sativa</i>	Transformation of IFN gene from Atlantic Salmon	Prevention of viral infections in fishes	Fukuzawa et al. (2010)
13	LL-37	<i>Hordeum vulgare</i>	Genetic transformation of chimeric LL-37	Molecular farming for production of therapeutic antimicrobial Peptide LL-37	Holásková et al. (2018)

Table 4

A summarized list of CRISPR-Cas mediated genome-editing events in plants for manipulating the metabolic pathways and/or biotechnological applications.

S. no.	Plant species	CRISPR-Cas system	Approach	Application	Reference
1	<i>Papaver somniferum</i>	CRISPR-Cas9 system	CRISPR-Cas9 mediated genome editing to knock-out 4'O-methyltransferase (gene which regulates the biosynthesis of benzyloisoquinoline alkaloids (BIAs).	Regulation of BIA metabolism and biosynthesis	Alagoz et al. (2016)
2	<i>Salvia miltiorrhiza</i>	CRISPR-Cas9 system	CRISPR-Cas9 mediated knock-out of diterpene synthase gene (SmCPS1)	Pathway elucidation of secondary metabolites, quality and yield improvement	Li et al. (2017)
3	<i>Glycine max</i>	CRISPR-Cas9 system	CRISPR-Cas9-mediated multiplex gene-editing of <i>GmF3H1</i> , <i>GmF3H2</i> and <i>GmFNSII-1</i> in soya bean plants and hairy roots	Increased Isoflavone content and plant tolerance to soyabean mosaic virus	Zhang et al. (2020)
4	<i>Solanum lycopersicum</i>	pYL-CRISPR-Cas9 system	pYL-CRISPR-Cas9 system knock-out of phytoene desaturase gene	Increased γ -aminobutyric acid levels	Li et al., 2018a, b
5	<i>S. lycopersicum</i>	CRISPR-Cas9 system	Targeted mutagenesis of the auto-inhibitory domain of <i>SIGAD2</i> and <i>SIGAD3</i> genes to increase Glutamate decarboxylase activity	Increased γ -aminobutyric acid levels	Nonaka et al. (2017)
6	<i>Arabidopsis thaliana</i>	CRISPR-Cas9 system	Targeting a methylated promoter to activate an silenced gene in <i>A. thaliana</i>	Tool for manipulation of silenced genes in plants	Lowder et al. (2015)
7	<i>Nicotiana benthamiana</i>	CRISPR-Cas9 system	Targeted chromosomal deletions in tobacco using <i>Agrobacterium tumefaciens</i> -mediated transient expression system	For precise and effective genome editing	Lowder et al. (2015)
8	<i>Oryza sativa</i>	CRISPR-Cas9 system	Targeted deletion and mutagenesis using multiplex	To validate efficiency of CRISPR-Cas9 in genome editing and compare OsU6 and OsU3 promoters	Lowder et al. (2015)
9	<i>Citrullus lanatus</i>	CRISPR-Cas9 system	Targeting CIPDS (phytoene desaturase) gene for targeted mutagenesis	Genome editing to know gene function and create agronomically important mutations in crops	Tian et al. (2017)
10	<i>Vitis vinifera</i>	CRISPR-Cas9 system	CRISPR-Cas9 mediated targeted mutation of phytoene desaturase (PDS) gene	Molecular breeding of <i>V. vinifera</i> , crop improvement	Nakajima et al. (2017)
11	<i>Cucumis sativus</i>	Cas9/sgrRNA system	Cas9/sgrRNA construct targeting of eIF4E (eukaryotic translation initiation factor 4E) gene	Plant improvement, Virus resistance in plants	Chandrasekaran et al. (2016)
12	<i>S. lycopersicum</i>	CRISPR-Cas9 system	CRISPR-Cas9 mediated induced somatic mutations into <i>SIL1A9</i> (a gene controlling parthenocarpy)	Rapid breeding of parthenocarpic tomato plants	Ueta et al. (2017)
13	<i>Malus domestica</i>	CRISPR-Cas9 system	CRISPR-Cas9 mediated targeted mutation of endogenous phytoene desaturase (PDS) gene	Improved molecular breeding to generate desired traits	Nishitani et al. (2016)
14	<i>Hordeum vulgare</i>	RNA-guided Cas9 system	RNA-guided Cas9 genome editing by targeting two copies of HvPM19 gene	Rapid characterization of gene function and crop improvement	Lawrenson et al. (2015)
15	<i>Brassica oleracea</i>	CRISPR-Cas9/sgrRNA system	CRISPR-Cas9/sgrRNA system mediated targeting of canker susceptibility gene <i>CsLOB1</i>	To produce disease-resistant citrus varieties	Jia et al. (2017)

**Fig. 3.** Computer-aided construction of the metabolic models in plants for production of antimicrobials/antibiotic reversal agents.

metabolic engineering have facilitated the creation of plant metabolic models to delineate the complex metabolic networks (Baghalian et al., 2014). The modeling of plant metabolic networks through computer-aided designing provides a deeper insight into the complexities of plant metabolism and facilitates its manipulation (Fig. 3). The construction of a metabolic model *via* systems biology includes essentially four steps, the *in silico* tools to establish the structure of plant metabolic network, the mathematical modeling and visualization to construct a comprehensive model, validation of the plant metabolic model by experimental, and *in silico* testing to remove errors (Klipp and Schaber, 2006; Zhang et al., 2010). A robust plant metabolic model should predict the future outcome of the experiments, and the model analysis/prediction potential in experimental analysis (Baghalian et al., 2014).

SynBio strategies have made good strides in exploring plant chassis for designing plant metabolic pathways and its outcome in heterologous systems. Despite gaps and challenges in the artificially created metabolic pathways, their feasibility may be predicted by advanced computational approaches (Küken and Nikoloski, 2019). Co-integration of *in silico* resources with SynBio have made considerable progress in modeling different plant parts for extraction purpose, besides a better prediction of outcomes in reconstituted pathways and refinement of plant metabolic networks (Küken and Nikoloski, 2019). On similar lines, computational tools for discovery of metabolic pathways have achieved success in plant chassis initiatives, pathway reconstitution in desired organism (Kotera and Goto, 2016) and identification of pathway designs beyond predictive methods (Medema et al., 2012). The graph-based, retrosynthesis-based and stoichiometry-based methods have aided in design and feasibility of metabolic pathways, attributed to database use and representation, algorithm searching and ranking of best metabolic pathway (Wang et al., 2017). Plant chassis and production of PNPs necessitates a methodical workflow employing computer aided design in metabolic pathway designing.

The progress in development of efficient and cost-effective AMPs may be considered a major contribution of computational strategies together with advanced design approaches. Tools in computational biology along with machine-learning, stochastic methods and empirical approaches contributed to peptide optimization and design AMP variants (Porto et al., 2012). Computer aided design of AMPs has co-integrated bioactivities and chemical parameters of AMP sequences information, facilitating prediction mode of antibacterial function of a candidate gene sequence, prior to its synthesis. While, the computational models, quantitative structure-activity relationship (QSAR) have enhanced optimization of AMP sequences with improved activities, the *de novo* design, genetic algorithms, and linguistic model holds prospects to enhance AMP automated design (Cardoso et al., 2020). The determinants necessary for antimicrobial functions can be identified by QSAR models and be used to design novel and effective AMPs (Lee et al., 2017). Haney et al. (2018) worked on discovery of peptides which act on bacterial biofilms, the QSAR model correlated the anti-biofilm potential of the variants with the descriptors and predicted the anti-biofilm activity of a peptide from a virtual library of 100,000 peptides. Further, the *de novo* design of AMPs aid in studying multiple AMP sequences and together with SynBio highlights a promising approach for creation of biomimetic antimicrobial polymers having therapeutic functions, mimicking AMPs (Woolfson et al., 2015). Effective drug candidates were developed employing *de novo* algorithms based strategies, a key example is omiganan (MBI-226), a peptide-based engineered AMP (Faccione et al., 2014). In another study, quorum sensing inhibitors (small molecules) were identified through computational approaches (from a database of drugs/natural products). SuperDrug and SuperNatural databases (Goede et al., 2005; Dunkel et al., 2006) were screened for 147 known drugs and natural products (based on 2-dimensional structural similarity). Molecular docking (Molegro Virtual Docker) was employed for screening QSI(s). Three compounds nifuroxazide, salicylic acid, and chlorzoxazone showed strong quorum sensing inhibition (QSI) gene expression and related phenotypic effects, and have good potential as

anti-pathogenic drugs (Yang et al., 2009). Structure-based screening employing computational approaches highlights an important method to search for novel antibacterial compounds.

Interestingly, several databases have been developed for prediction and designing of AMP sequences. One such database, the Biofilm-active AMP database (BaAMPs, <http://www.baamps.it>, Di Luca et al., 2015) comprises the manually extracted and curated information regarding AMP sequence, chemical modifications in target organisms, methodologies and peptide concentrations involved during biofilm inhibition/reduction. Another database, Cybase (Cyclic protein database, <http://research1t.imb.uq.edu.au/cybase>, Wang et al., 2007) contain tools for visualization, characterization and engineering of cyclic proteins. The CAMPR3 (Collection of Anti-Microbial Peptides) database (<http://www.camp.bicnirrh.res.in/>, Waghu et al., 2016) comprises the information on sequence signatures conserved in AMPs, along with the provision of sequence alignment, pattern creation and Hidden Markov Model based searching. Another one, the Antimicrobial Peptide Database (APD, <http://aps.unmc.edu/AP/main.html>, Wang and Wang, 2004) comprises the information for peptides along with their statistical data along with the interactive interface for peptide query, predictions and design. The YADAMP (yet another database of antimicrobial peptides, <http://yadamp.unisa.it/about.aspx>, Piotto et al., 2012) database is developed to facilitate the access to AMP information with a precise focus on short alpha helix peptides interacting with cell membrane. The AMSDB database (<http://www.bbcm.univ.trieste.it/~tossi/amsdb.html>, (Tossi et al., 2000) comprises the data of annotated AMP sequences. Recent reports highlighted the advantages of computer-aided design in aiding antimicrobial discovery and development, like enhanced antimicrobial function and activity of Cefuroxime Axetil/Cyclodextrin System through computer-aided design (Mizera et al., 2019), using computational tools to design lipopeptide as novel antimicrobial drugs (Eswari et al., 2019), computer-aided identification of recognized drugs as *Pseudomonas aeruginosa* quorum-sensing inhibitors (Yang et al., 2009), designing of novel peptides for broad-spectrum antimicrobial activity through computational tools (Liu et al., 2015), and designing effective peptide antibiotics through *in silico* optimization (Porto et al., 2018). A better understanding of antimicrobials and target interaction using plant chassis may be achieved through co-integration of laboratory data with computational resources. The advents in data-mining methods and biological databases is expected to further assist in the discovery and development of effective antimicrobial agents.

Though computer-aided design of antibiotics is historically focused on peptides and proteins with significant success (Lee et al., 2017); however, recent approaches in discovery and designing of novel antimicrobials have explored other natural products as well. For instance, Dai et al. (2018) reported the synthesis and designing of new 3,9-disubstituted eudistomin U derivatives employing computer-aided drug discovery. Further, employing whole-genome sequencing for antibiotic susceptibility testing (WGS-AST) represents an emerging technique which predicts AMR in bacterial genomes by the presence or absence of single-nucleotide polymorphisms (SNPs) and genes in the bacterial genomes (Su et al., 2019). The WGS-AST has the potential to detect antibiotic resistant phenotypes in the entire genome simultaneously and screening of phenotypes (because of multiple loci) (Su et al., 2019). The machine learning (ML) models were employed to test mutations subsets to determine the predictability of the model (Yang et al., 2018). In addition to the bacterial genomes, progress has been made in employing WGS for prediction of AMR in fungi, viruses and eukaryotic parasites (Depledge et al., 2014; Lockhart et al., 2017).

5.4. Metabolic engineering or introduction of synthetic pathways in plants

A new era in antimicrobial discovery foresees implementation of existing and upcoming SynBio strategies to remodel natural systems and re-program biological networks (Cravens et al., 2019). Plants cell factories have been explored for production of pharmaceutical metabolites,

including paclitaxel (Wani et al., 1971; Fett-Neto et al., 1992), camptothecin (Wall et al., 1966; Sakato et al., 1974) and artemisinin (Nair et al., 1986), among others, refer to Table 2. However, the production of PNPs in meager quantities, plant complexity and extended life-cycle accounts are some key challenges. Metabolic engineering of plant systems highlights a powerful strategy in plant chassis for generation of novel metabolites, considering an expanding toolkit of biosynthetic entities. With the availability of whole genome sequences of several plant species and accessibility to the “big-data”, metabolic engineering projects a cost-effective and scalable option for enhanced production of desired metabolites (O'Connor, 2015). The widespread recognition of the fact that secondary metabolites represent “privileged scaffolds” during evolutionary course, help projects them as ideal “drug-leads” in drug discovery process. The pathway engineering for PNP production offers multi-faceted benefits, in terms of structural modifications/ generation of novel entities, improved biological functions of host organism (O'Connor, 2015). With the advances in molecular biology, introduction of the entire gene cluster, reconstituting the entire pathway or even the transfer of an entire metabolic pathway into heterologous systems are being attempted. For instance, Fuentes et al. (2016) optimized a new plant SynBio approach, the combinatorial supertransformation of transplastomic recipient lines (COSTREL), and used it for introducing the complete pathway for artemisinic acid, the precursor of artemisinin, into tobacco. The authors successfully engineered the pathway, and the resultant COSTREL lines could produce over 120 mg artemisinic acid per kilogram biomass (Fuentes et al., 2016). Further, Lu et al. (2017) introduced a synthetic operon encoding biosynthetic pathway components for astaxanthin in the plastids of tobacco, and the transplastomic plants accumulated astaxanthin up to 1% of the plant dry mass. The methods to incorporate genes into the desired location of host genome would ensure stable gene expression. Furthermore, regulation of protein expression may be achieved by characterization of regulatory elements (promoters), playing a substantial role in metabolic engineering and reconstitution of complex pathways in suitable host by DNA blocks assemblage (O'Connor, 2015). Another major goal of plant chassis accounts for construction of plant metabolic pathways for enhanced PNP production, which are cumbersome to produce by natural plant systems (Fesenko and Edwards, 2014).

Various other strategies were also adopted in plant metabolic engineering such as use of CRISPR-Cas for biosynthetic pathway manipulations (Zhou et al., 2018), transcriptional regulation for multi-enzymatic steps in metabolic pathways (Grotewold, 2008), enzyme engineering to increase or decrease the targeted metabolite (Nonaka et al., 2017), enzyme inhibition for enhanced metabolite production (Alagoz et al., 2016), inhibition of branch pathway for removing competitive pathways and redirecting the substrate to particular metabolite (Lv et al., 2016), and improving precursor supply (Wu et al., 2015) among others. These strategies in plant metabolic engineering have opened new avenues in plant chassis towards production of high-value metabolites and genome editing.

5.5. Metabolic flux modulations

In plant metabolic network, an understanding of the metabolic phenotypes (measurement of metabolic flux) and flux distribution is essential (Ferne et al., 2005) for the analysis of carbon efficiency and selection of targets for metabolic engineering (Shih and Morgan, 2020). Additionally, the construction of metabolic networks and flux prediction provides a platform for gaining insight about the network processes. Plant metabolic flux is often regulated by a rate-limiting enzyme that affects the pathway rate, studied by metabolic control analysis. In plant secondary metabolism, there are several rationales for quantifying metabolic fluxes: the need to quantify key precursors of secondary metabolites, to understand and quantify the metabolic flux splits at the branch point, and the identification of alternate routes (undesired flux) leading to side products. Therefore, the analysis of metabolic flux and its

regulation may provide a systematic approach to understand plant secondary metabolism and networks. In addition, the metabolic flux in primary metabolism provides precursor supply for the biosynthesis of secondary metabolites and quantification of metabolic flux would provide a better understanding of the environmental influence on primary metabolism and synthesis of precursors for secondary metabolism (Shih and Morgan, 2020).

While in metabolic systems, flux control refers to its alteration by changes in enzyme activity, flux-regulation can be defined in terms of how external changes influence any metabolic process (Fell, 1997). The regulatory enzymes control the intermediate concentrations of metabolites to limit the effect of external changes, maintaining metabolic homeostasis. A comprehensive knowledge about metabolic flux and its modulation may thus facilitate plant metabolic engineering, especially in case of high-titer products. In this manner, high-value metabolites may be produced through modulation of metabolic fluxes employing SynBio strategies. Another key aspect for consideration in metabolic flux modulation is the highly branched nature of metabolic pathways and the role of multiple enzymes in flux control (Morandini, 2013). Therefore, an approach for flux modulation would focus on multiple sites in the pathways (Kruger and Ratcliffe, 2012). The modulation of metabolic flux via engineering would focus on the best candidate gene(s) (Lee et al., 2011a) and an in-depth understanding of regulatory networks in plant metabolism (Stitt et al., 2010), as depicted in Fig. 4. Studies on metabolic flux modulation and its analysis enable the targeted modification of plant metabolic pathways and plant chassis for enhanced production of targeted metabolite. In an investigation conducted by Horbal et al. (2018), the increase in botryomycin production was not only attributed to the changes in metabolic fluxes but also to the synthetic promoters used. Further, authors decoupled the gene expression from native gene regulation (e.g. feedback loops that downregulate gene expression via native promoters) (Horbal et al., 2018). However, more studies are needed to explore this approach for better understandings.

5.6. RNAi in plants for production of phyto-antimicrobials

As an evolutionary conserved mechanism, RNA interference (RNAi) or silencing mechanisms in plants were uncovered as small interfering (si)-RNA, transcriptional RNA-directed DNA methylation (RdDM) and post-transcriptional micro (mi)-RNA pathways (Guo et al., 2016) and formed the basis of artificial gene-silencing strategies in plants. The inherent RNA silencing mechanism in plants is activated by double-stranded RNA (dsRNA) comprising Dicer-like (DCL) and Argonaute (AGO) proteins (Baulcombe, 2004; Meister and Tuschl, 2004). The mechanism of RNA-silencing comprise of a Dicer protein which process the dsRNA into small RNA (20–24 nucleotides) duplex with overhangs at both ends. An RNA-induced silencing complex (RISC) is formed by incorporating one small RNA strand into AGO protein. The RISC complex is guided to ssRNA (by the small RNA) and AGO protein cleaves RNA at 10–11 nucleotide in the central region (Baulcombe, 2004; Meister and Tuschl, 2004). The RNA silencing mechanism in plants have evolved for performing different functions and have diversified into multiple overlapping pathways including RNA-directed DNA methylation pathway, microRNA (miRNA) pathway, exogenic RNA silencing pathway and trans-acting small interfering RNA (tasiRNA) pathway (Guo et al., 2016).

To address the growing concern of AMR, employing RNA silencing approaches, the CRISPR-RNA (crRNA) for prokaryotes and siRNA for eukaryotes, with a complementary sequence to drug-resistance-encoding mRNA highlights a prospective approach (Hale et al., 2009; Eidem et al., 2012). The RNAi has achieved great attention in antimicrobial therapies (specifically in drug-resistant infections) and development of antimicrobial agents, as genetic engineering tool (McGinnis, 2010), defense mechanism (Obbard et al., 2009) and gene regulation (Eamens et al., 2008). It defines a revolutionary strategy in combinational therapies to counter drug-resistance by knocking down the

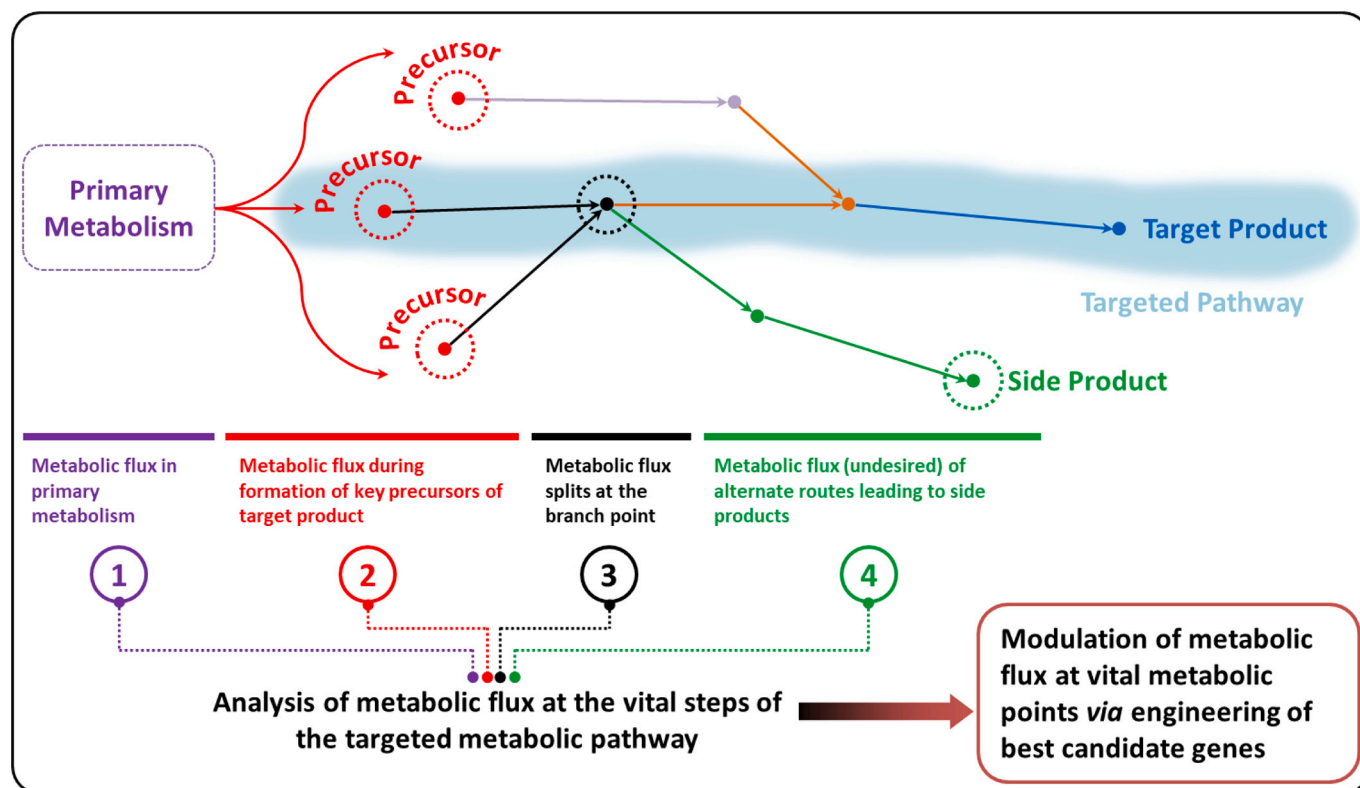


Fig. 4. Hypothetical metabolic pathway displaying the vital steps for metabolic flux analysis of the pathway. The represented vital steps can be aimed for metabolic flux modulation.

metabolic pathways with specificities at the genome levels (Edson and Kwon, 2014).

Genetic engineering of crRNA sequences, which would promote AMR-enablers silencing, is required. The synthetic crRNAs, exhibits high scalability and have excellent prospects in biomedical applications (Reynolds et al., 2004). Plant chassis for antimicrobials employing RNAi based systems are emerging. Studies on gene silencing in plant systems for improving resistance to diseases (Galambos et al., 2013), quality improvement and stress tolerance in horticultural crops (Parmar et al., 2017), engineering virus resistance in plants via RNA silencing (Baulcombe, 2004), suggested that antiviral RNAi constitutes key defense process in plants for plant protection. The technology has been useful in silencing the drug-resistance enablers through antimicrobial RNA delivery (Hale et al., 2008). In a relevant example, crRNA acting against LmrA or MsrA counters drug-resistance of *E. coli* and MSRA strain to antibiotics (Van Bambeke et al., 2003). In another RNAi study targeting AMR, the crRNA delivered in *Synechocystis* sp. degraded the target mRNA (Abe et al., 2014). Fig. 5 postulates how RNAi and CRISPR-Cas approaches can be effectively used for the production of targeted phyto-antimicrobials via plant genome editing.

Interestingly, RNAi has also been coupled with nanotechnology and is an emergent approach for combating AMR. Nanoparticles have to be covalently linked with cationic peptides (for antimicrobial RNA delivery), and form complex with RNA sequence (Paunesku et al., 2003). Recent advances in RNAi based antimicrobial therapies aims to introduce nano-scale formulations (nano-antibiotics) for administration of therapies and prospects of sensitizing drug-resistant strains to antimicrobial therapies (Edson and Kwon, 2014). A key example showed that siRNA complex (combination of cationic peptides or cationic polymers), carriers of antimicrobials and inorganic nanoparticles may be efficient in removing drug-resistant microbes (Xu and Zhou, 2008). The combinational approach also aims at photothermal therapy and RNAi as a multi-modal antimicrobial therapy and specific designing of siRNA and

crRNA against drug-resistant enablers may be a prospective approach in achieving safe and effective antimicrobial activity and minimizing the AMR-emergence (Grissa et al., 2007). While the prospects of RNAi-based therapies are immense, however, consequences due to off-targets formed by crRNA targeting microbes in eukaryotic cells. These off-targets may appear because of a nucleotide mismatch of 2 or more when generating specific sequences which results in undesired chromosomal translocation in eukaryotic host cell (Fu et al., 2013; Cho et al., 2014b). A possible way to reduce off-targets is by identification of unique sequences specific to drug-resistant microbes (Cho et al., 2014b) and designing particular crRNA (protected by engineered antimicrobial RNA carriers). Though RNA silencing technologies still projects certain limitations like knowledge about whether the technology may switch off endogenous gene promoters (most likely resistant to siRNA-directed transcriptional silencing) in plants, limits their use. However, a prospective strategy co-integrating PTGS with other silencing mechanisms would provide better outcome (Guo et al., 2016). The RNA silencing mechanisms needs an in-depth understanding and improvement to counter potential limitations.

5.7. Plant genome editing/engineering

The availability of big-data and knowledge about the biological systems as well as high-throughput technologies have shifted the attention on regulation of gene-networks and development of promising approaches to address AMR (Singh, 2015). Recent gene editing tools including Transcription Activator-Like Effector Nucleases (TALENs), Zinc-Finger Nucleases (ZFNs) and CRISPR-Cas are opening new avenues in genome editing/engineering (Dangi et al., 2018). CRISPR-Cas defines an immune system which protects archaea and bacteria from pathogenic microbes. In this approach, small sequences designated as spacers (foreign DNA/RNA elements) are incorporated in the bacterial genome (CRISPR loci) and the Cas protein recognizes and destroys the invading

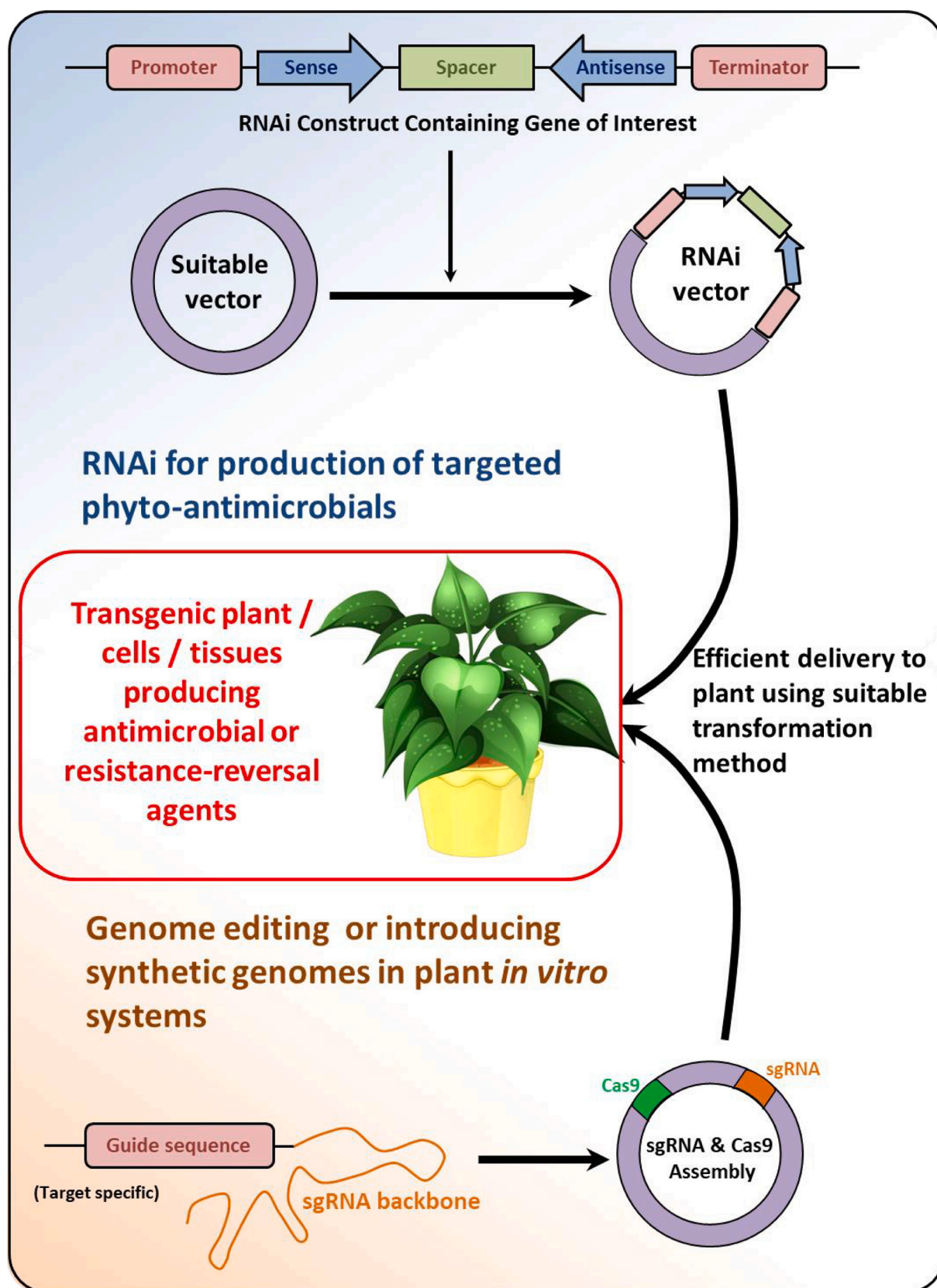


Fig. 5. RNAi and CRISPR-Cas mediated plant genome editing approaches for the production of desired phyto-antimicrobials.

pathogenic nucleic acids through CRISPR loci. The CRISPR-Cas systems (genome editing/ engineering tools) are classified into two classes, type-1 has a complex framework and the Cas protein functions in foreign DNA recognition and its cleavage, while class 2 systems are relatively simple, with DNA recognition and cleavage by a single multi-domain enzyme (Singh et al., 2017). The type-2 system is employed more frequently in genome editing.

CRISPR-Cas system represents a programmable tool to precisely inhibit the invading microbe and thus opening new possibilities in tackling drug-resistant infections (Beisel et al., 2014). de la Fuente-Núñez and Lu (2017) designed CRISPR-Cas systems as antimicrobials to remove drug-resistant microbes, specifically targeting genes in biofilm formation, antibiotic resistance and genes involved in virulence. Significant advances in employing CRISPR-Cas system highlights massive potential to re-sensitize AMR bacteria or sequence specific killing. However, the focus of this review is to present and discuss the possibilities of using tools like CRISPR-Cas for editing plant genomes for altered production of potent phytomolecules, therefore, genome editing of drug-resistant microbes either for killing or re-sensitizing them is not discussed further.

Though CRISPR-Cas mediated strategies for genome-editing or creation of synthetic genomes has mostly been explored in microbes, plant systems are also emerging as sound platforms. Recent advances in plant genome editing and transcriptional-regulation via CRISPR-Cas9 system presents a revolutionary strategy. The CRISPR-Cas system highlights a prospective approach to activate endogenous gene transcription in plants (Lowder et al., 2015), leading to a rapid and streamlined use of this approach in plant research. A molecular toolbox was developed by Lowder et al. (2015) for efficient use of CRISPR-Cas comprising information about the reagents; protocol for assembly of DNA constructs in plants and has been tested in *Oryza sativa*, *Nicotiana benthamiana* and *Arabidopsis thaliana* (Lowder et al., 2015). Similarly, CRISPR-Cas12a is gaining momentum as an efficient genome editing tool because of its versatility, simplified features and its abilities to generate double-strand breaks with staggered ends (Bandyopadhyay et al., 2020). Genome editing/engineering in plants employing CRISPR-Cas approach has shown good leads in producing desired PNPs including phyto-antimicrobials. With emphasis on the plant secondary metabolism, these approaches have mainly been used to explore the biosynthetic perspective of plant systems via altering the metabolic fluxes for production of targeted PNPs (Marchev et al., 2020). An attempt towards CRISPR-Cas9 mediated plant chassis aimed at mutagenesis of rosmarinic acid synthase (RAS) and regulation of phenol biosynthetic pathway in *Salvia miltiorrhiza* (Zhou et al., 2018). The designed guide RNA precisely edited rosmarinic acid synthase gene (SmRAS) gene. The generated mutants from transgenic hairy root were obtained in sgRNA driven by *Arabidopsis* U6 promoter, while mutants were absent in sgRNA driven by the rice U3 promoter. The expression analysis showed a decreased expression of RAS and phenol content in the engineered hairy root lines (Zhou et al., 2018). The study showed that CRISPR-Cas9 system may be effectively utilized for genome editing in *S. miltiorrhiza* highlighting potential in regulation of natural product pathways and enhanced PNP production. Zhang et al. (2020) employed a multiplex CRISPR-Cas9 for metabolic engineering of *Glycine max*, which resulted into an increased isoflavone content and resistance to soybean mosaic virus. In another significant study, the metabolic engineering of *S. lycopersicum* was performed employing pYL-CRISPR-Cas9 system to target phytoene desaturase gene, and the resultant genome-edited plants showed higher γ -aminobutyric acid (GABA) contents over their wild-type counterparts (Li et al., 2018a, 2018b). Similarly, Li et al. (2017) demonstrated CRISPR-Cas driven knocking-out of diterpene synthase gene (SmCPS1) in *Salvia miltiorrhiza*, which blocked the geranyl-geranyl diphosphate and disrupted the tanshinone biosynthesis. In another attempt, a 3-hydroxy-N-methylcoclaurine 4-O-methyltransferase (4'OMT2) gene was knocked out by CRISPR-Cas9 genomic-editing in *Papaver somniferum*, which resulted in altered production of benzyloquinoline alkaloids

(morphine, noscapine, papaverin) (Alagoz et al., 2016). Some of these PNPs are reported to possess antimicrobial potentials (Ekambaram et al., 2016; Feng et al., 2019). These findings suggest that though this area is still in its infancy and far from its potential exploration, the CRISPR-Cas system may be precisely and effectively used for plant genome editing and alteration of metabolite content in plants for producing desired phyto-antimicrobials in plant chassis to counter AMR. Some key studies employing CRISPR-Cas systems for editing biological genomes for antimicrobial production are presented in Table 4.

6. Conclusion and future outlook

Plant chassis for production of potent and novel phyto-antimicrobials offer a prospective approach to address the emerging threat of AMR. Recent advances in high-throughput technologies have facilitated the discovery and characterization of PNPs as “potent antimicrobials” with expanding SynBio toolkit. Recent advances in metabolic engineering, genome editing/engineering and strategies in computer-aided design and target-identification are opening new avenues in tailoring plant hosts as production platforms for novel, cost-effective and sustainable antimicrobials. The ever-increasing data-set on plant genomes and transcriptomes have facilitated the identification of primary metabolic pathways/enzymes and the specialized ones and the unique pathways present in specific lineages (Maeda, 2019). However, it is equally important to optimize plant metabolic pathways in terms of precursor supply for production of targeted metabolites, and metabolic flux modulation seems a potential strategy in this direction. Tailoring plant hosts for the production of specific metabolites may be envisioned through the use of plant chassis for expression of natural enzyme variants of primary plant pathways and identifying mutations of natural enzyme variations. Unraveling the primary metabolic diversity in diverse plant lineages together with prior understanding and scientific technologies would be the key towards sustainable production of natural products and a bio-based economy. With a focus on newer and efficient SynBio strategies, plant chassis for production of novel antimicrobials forms the key theme of this article. The plant hosts as “production platforms” for antimicrobials highlight an emerging and sound strategy to counter the challenges associated with AMR in microorganisms. Though it is far from its potential exploration, and invites more investigations for establishing, advancing and exploring plant SynBio approaches for production of high-value phyto-antimicrobials.

Declaration of competing interest

None to declare.

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